

PHYSICS GRADE 10: ELECTROSTATICS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.4 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 3.0: Electricity and Magnetism
Sub-Strand	Sub-Strand 3.4: Electrostatics
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Electric charges: positive and negative charges, charge on atoms and subatomic particles – Charging by friction: transfer of electrons between insulators (rubbed balloon, polythene, perspex, ebonite, glass, silk, fur) – Conductors and insulators: free electrons, bound electrons, examples in Kenyan homes and industry – Charging by contact and by induction: using an electroscope, earthing – Law of electrostatics: like charges repel, unlike charges attract; charging and attraction of neutral objects – Distribution of charge on conductors: charge on the outer surface, effect of shape, action of points (corona discharge) – Detecting and measuring charge: the gold-leaf electroscope, charge detection, testing the sign of charge, discharging – Applications of electrostatics: lightning conductors, photocopiers, electrostatic precipitators in cement factories, paint spraying, crop spraying – Hazards of static electricity and safety: lightning, fuel tanker earthing, static in dry environments
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - Describe the origin of electric charges and how materials become charged - Distinguish between conductors and insulators of electricity - Investigate charging of conductors by contact and induction - Explain the distribution of charge on a conductor - Detect and measure electric charge using a gold-leaf electroscope - Appreciate the application of electrostatics in everyday life
Core Competencies	– Critical thinking and problem solving: learners analyse evidence from charging activities to explain attractions and repulsions and to predict outcomes in new situations – Communication and collaboration: learners work in groups to carry out investigations, share observations, and defend explanations using evidence – Creativity and imagination: learners design simple electroscopes and devices from locally available materials – Digital literacy: learners use simulations to visualise the movement of electrons during charging – Learning to learn: learners revisit the Driving Question Board and revise their models as understanding grows
Core Values	– Responsibility: learners handle electrical materials and equipment safely and care for shared apparatus – Integrity: learners report observations honestly even when results differ from predictions

	– Unity: learners cooperate in groups to carry out investigations and build shared models – Respect: learners listen to and value the ideas of others during discussions
Science & Engineering Practices	– Asking questions and defining problems: learners generate questions about the balloon, hair and paper on the Driving Question Board – Developing and using models: learners build and revise a particle model of electron transfer – Planning and carrying out investigations: learners test materials for charging and conduction – Analysing and interpreting data: learners interpret electrostatic observations – Constructing explanations: learners explain attraction of neutral objects using charge ideas – Engaging in argument from evidence: learners debate explanations of charge distribution
Pertinent & Contemporary Issues (PCIs)	– Safety and security: learners learn lightning safety and the safe handling of fuel and electrical equipment – Environmental education: electrostatic precipitators reduce dust pollution from cement factories – Life skills: learners apply decision making when protecting themselves and property from lightning – Education for sustainable development: learners relate electrostatic technologies to efficient use of resources, such as crop spraying
Career Connections	– Electrical engineer: designs safe electrical systems and lightning protection for buildings in Nairobi and Mombasa – Electronics technician: handles static-sensitive components when repairing phones and computers – Meteorologist: studies lightning and thunderstorms, such as those in the Lake Victoria basin – Industrial or chemical engineer: applies electrostatic precipitators and spray painting in factories – Agricultural technologist: uses electrostatic crop sprayers on farms in Kericho and the Rift Valley – Photocopier technician: services machines that rely on electrostatic charge
Focus for Lessons	This unit uses the puzzling behaviour of a rubbed balloon to drive learners to investigate where electric charges come from and how they behave. Learners predict, observe, explain, record questions on the Driving Question Board, and progressively revise a particle model of charge across nine lessons. The unit connects to Kenyan contexts such as lightning in Kisumu, cement dust control and fuel tanker safety.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? KICD KEY INQUIRY QUESTIONS: 1. Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? 2. What are electric charges? 3. Why do some materials conduct electricity while others don't? 4. How can we charge conductors? 5. Where does charge go on a conductor? 6. How can we detect and measure electric charge? 7. How is electrostatics used in everyday life?
Anchoring Phenomenon	When you rub a balloon on your hair, your hair stands up and follows the balloon as you pull it away. The same balloon then sticks to a classroom wall for several minutes without glue or tape, and small pieces of paper jump toward it without touching it. On a dry day in Nairobi, a learner may also feel a small shock after combing hair or removing a woollen jersey. It is puzzling because nothing visible connects the balloon to the hair, wall or paper, and the balloon was never magnetic or plugged into electricity. What causes these mysterious forces? Where do the charges come from? Why does the balloon sometimes attract and sometimes

	repel? Students observe these surprising attractions and repulsions and wonder about the invisible forces at work.
Storyline Thread	Lesson 1: Anchoring phenomenon. Students rub balloons on hair, stick them to walls and lift paper pieces; they record observations and questions on the Driving Question Board and draw initial models. Lesson 2: What are electric charges? Students study atoms, protons, electrons and neutrons, and test the rule that like charges repel and unlike charges attract using rubbed rods and strips. Lesson 3: Charging by friction. Students rub different materials (polythene, perspex, ebonite, glass, silk, fur), identify the sign of charge, and model electron transfer to explain the balloon and hair. Lesson 4: Conductors and insulators. Students test materials such as copper, aluminium, plastic, wood and metal spoons, and explain differences using free electrons. Lesson 5: Charging by contact. Students charge conductors by touching with a charged rod, and explain the sign of charge gained using electron movement. Lesson 6: Charging by induction and earthing. Students charge conductors without touching them and explain how this clarifies why neutral paper is attracted to the balloon. Lesson 7: Distribution of charge on conductors. Students investigate charge on outer surfaces and at points, and explain sparks and lightning conductors. Lesson 8: Detecting and measuring charge. Students use a gold-leaf electroscope to detect charge, find its sign and discharge it, and relate leaf divergence to amount of charge. Lesson 9: Applications and consolidation. Students analyse applications such as lightning conductors, photocopiers, electrostatic precipitators and fuel tanker earthing, revise their final model, and answer the main driving question using the Driving Question Board.
Supporting Phenomena	– A plastic comb rubbed on dry hair picks up tiny pieces of paper or sukuma wiki leaf fragments – Lightning flashes during thunderstorms over Kisumu and Lake Victoria, and tall buildings are fitted with lightning conductors – Fuel tankers delivering petrol in Mombasa trail an earthing chain to prevent sparks – Cement factories use electrostatic precipitators so that chimneys release less dust – A gold-leaf electroscope leaf rises when a charged rod is brought near the cap, without touching it

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute periods)): The Mysterious Balloon: What Invisible Force Is at Work?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon of the mysterious balloon, so that learners observe and describe electrostatic attraction and repulsion without explaining it, record their questions on the Driving Question Board (DQB), and draw initial models that will be revised in every later lesson.
Knowledge	Learners describe what a rubbed balloon does to hair, a wall and paper pieces, and what an unrubbed balloon does. They distinguish attraction from repulsion, and note that the effect is weaker on humid days (Mombasa, about 80%) than on dry days (Nairobi, about 35%).
Skills	Predicting, observing and recording carefully, asking testable questions (NGSS SEP 1: Asking Questions), developing and using models (NGSS SEP 2), planning a fair comparison with a control (unrubbed balloon), and communicating observations in talk and sketches.
Attitudes	Curiosity about invisible forces, willingness to say 'I do not know yet', respect for classmates' ideas, safe and careful handling of materials, and honesty in recording what is actually seen rather than what is expected.
Key Inquiry Question	Where do electric charges come from? Why does a rubbed balloon attract or repel other objects?
Purpose in Storyline	This is the opening lesson. It creates a shared experience and a need to know. Learners use only prior knowledge of magnets and everyday experiences such as combing hair. Explanations are deliberately withheld. Their questions populate the DQB, which drives Lessons 2 to 9, and their initial models are revised after every lesson.
Safety Notes	Use only dry, inflated latex balloons and check for latex allergy beforehand (use a polythene bag or another material for affected learners). Balloons must not be put in the mouth. Burst balloons are picked up at once to avoid choking. Keep thread lengths short. Handle scissors carefully when tearing or cutting paper. Rubbing is done gently, and no one pulls another learner's hair. Teacher demonstrations are done in dry conditions.

B. LESSON OVERVIEW

Mr Mwangi opens by asking learners about a time they felt a small shock or saw hair stand up after combing on a dry Nairobi morning. Learners predict what a balloon rubbed on hair will do, then work in groups of four to observe the phenomenon: 20 strokes of the balloon on dry hair, hair rising and following the balloon up to about 5 cm, the balloon sticking to the painted wall for about 3 minutes, torn paper pieces (about 1 cm x 1 cm) jumping up when the balloon is held about 2 cm above them, and two rubbed balloons on threads pushing apart. An unrubbed balloon serves as the control and does nothing. Learners then compare what they saw with Amina's report from Mombasa (humid, about 80%, it hardly works) and consider the Nairobi dry-day figure (about 35%). They describe the patterns with words such as attract and repel but do not explain them. Learners compare the effect with what they know about magnets, build the Driving Question Board (DQB) from their own questions, and draw initial particle-level or cause-and-effect models of the balloon, hair, wall and paper. These are kept for revision in later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall everyday experiences: Wanjiru notices her	Say: 'Think about the last time you combed your hair on a dry	Predict-and-Reason (Think-Pair-Share): writing a prediction and a	Check that at least 80% of learners wrote a prediction for all

 VIDEO: Pith ball electroscope Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Balloons and Static Electricity Source: PhET Interactive Simulations (English)  Search ARES for similar readings	hair stands up after combing on dry mornings in Nairobi, and others describe the small shock from a woollen jersey. Mr Mwangi shows a latex balloon. Individually, learners write and sketch predictions in their notebooks: (1) What will happen to my hair if I rub the balloon on it 20 times and pull it away slowly? (2) What will happen if I put the rubbed balloon against the wall? (3) What will happen if I hold it about 2 cm above small pieces of paper? (4) Will an unrubbed balloon do the same? They compare with a partner and then share with the group. They also state whether the balloon is like a magnet and give a reason.	morning. What did you notice? Turn to your partner and tell them.' Allow 1 minute for talk, then take WAIT TIME of 10 seconds before calling on anyone. Cold-call 3 learners, including Wanjiru, for their experiences. Hold up the balloon and say: 'I am not going to tell you what happens. I want your best prediction, and I want to hear your reasoning, not just your answer.' Give 3 minutes of silent writing, then WAIT TIME of 10 seconds before asking: 'Who predicts the balloon will behave like a magnet? Who disagrees? What makes you say that?' Cold-call 4 more learners to share and record predictions on the board in a tally. Do NOT confirm or correct any prediction. Say: 'All ideas are welcome today. We will test them.'	reason before seeing anything commits learners to an idea, so the later surprise becomes a real conflict between expectation and evidence. It also surfaces prior knowledge of magnets and combing hair that the teacher can build on in later lessons.	four questions plus a reason. Look for ideas such as 'it is magnetic', 'the hair is sticky', 'static' or 'nothing will happen'. Note which learners refer to magnets and which refer to everyday experiences. Record in the tally how many predict attraction, repulsion or no effect. Do not grade for correctness.
Observe Phase  VIDEO: Electric forces Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Balloons and Static Electricity Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Station 1, Hair: Each group of four rubs a dry balloon on the hair of one willing member for 20 strokes (about 10 seconds), then slowly pulls it away. They record what the hair does and the greatest distance (about 5 cm) at which hair strands follow the balloon. Station 2, Wall: They press the rubbed balloon against the painted wall, let go, and time how long it stays (about 3 minutes, range 2 to 5 minutes). Station 3, Paper: They tear paper into pieces of about 1 cm x 1 cm on a desk, hold the rubbed balloon about 2 cm above them, and record what happens, including pieces that jump up and then fall off again after touching the balloon. Station 4, Two balloons: They rub two balloons on hair, hang them from separate threads, bring them near each other, and record that they push apart. Control: An unrubbed	Before starting say: 'Your job right now is to be a careful observer. Write what you SEE, not why you think it happens.' Circulate and ask each group: 'What exactly did the hair do? How far from the balloon was it still moving?' and 'What did the unrubbed balloon do, and why do we test it?' If a group says 'because of static', respond: 'Static is a name. Tell me what you actually saw.' Remind all groups to use dry conditions and pick up burst balloons. After 20 minutes, stop and say: 'Hands down, pencils down. Look at your table.' WAIT TIME of 10 seconds. Cold-call 4 groups to report one observation each: hair, wall, paper, two balloons. Ask Otieno to describe the sticking balloon. Then show or read the humid-day data and say: 'Amina says it never works at her coastal home in the rainy season. What does our table	Structured Observation with a Control (NGSS SEP 3: Planning and Carrying Out Investigations): separating 'what I did', 'what I saw' and 'attract/repel/no effect' lets learners describe patterns without explaining them. The unrubbed balloon shows that rubbing is the cause of the effects, and the Nairobi and Mombasa data show that conditions matter, so learners see that the effect is real, repeatable and dependent on conditions.	Observe whether tables contain measurable details: about 5 cm for hair, about 3 minutes for the wall, about 2 cm for paper, balloons pushing apart, and no effect from the unrubbed balloon. Indicators of success: learners write what they saw without explanations, use the words attract and repel correctly, and record the control result. Note groups that report that paper pieces fall off after touching the balloon, which is a detail to return to in Lesson 5. Ask each learner to state one difference between dry and humid days using the data.

	balloon is tried on hair, wall and paper: no effect. They record all results in an observation table with columns 'What I did', 'What I saw', 'Attract / Repel / No effect'. Mr Mwangi then shows data: on a humid day in Mombasa (about 80% relative humidity) the balloon sticks for under 30 seconds or not at all, and in Nairobi's dry season (about 35%) a small shock is felt after removing a woollen jersey, with a faint crackle heard in a dark room.	tell us about conditions?' WAIT TIME of 15 seconds before taking responses from 3 learners. Ask Kipchoge to share his question about whether the shock is like lightning, and write it down for the DQB without answering it.		
Explain Phase  VIDEO: Pith ball electroscope Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners work in the same groups to sort their observations into two categories on a chart: 'Things that came together (attract)' and 'Things that moved apart (repel)'. They note which observations do not fit easily into either, such as the paper pieces that jump up and then fall off. Each group writes a 'we notice' statement (for example, 'The rubbed balloon pulls on hair, wall and paper without touching them, but the unrubbed balloon does not') and a 'we wonder' statement. They then compare the balloon with a magnet from prior knowledge, listing what is the same (pulls without touching) and what is different (does not need a metal object, pushes balloons apart, effect fades, depends on rubbing and weather). They share ideas about what could be happening but are told to treat them as hunches that will be tested, not final answers.	Say: 'We are not explaining yet. We are describing patterns. Which things came together? Which moved apart? What do you notice?' Give 8 minutes for group sorting. Circulate and ask: 'How do you know the balloon pulled the paper and the paper did not just move by itself?' (the unrubbed balloon control). Then ask: 'Is this a magnet? Think about the wall and the paper. What is the same and what is different?' WAIT TIME of 15 seconds. Cold-call 4 learners, one at a time, and ask each to build on the previous answer: 'Do you agree, and what would you add?' Record 'notice' statements on the board. If a learner proposes an explanation (for instance, 'tiny charges'), say: 'That is an interesting hunch. Write it on a sticky note for our DQB and we will test it over the coming lessons.' Do NOT confirm. Close with: 'What is the one thing we are all sure about, based on evidence?' Take 2 responses.	Notice-and-Wonder with Pattern Sorting (Constructing Explanations, NGSS SEP 6): sorting observations into attract and repel separates evidence from interpretation and builds a shared vocabulary. Comparing with magnets uses prior knowledge, then shows its limits, which creates productive uncertainty rather than an early, shallow answer.	Check that each group's chart separates observations from ideas and includes at least three entries in each category. Listen for correct use of 'attract' and 'repel' and for statements supported by evidence, such as the control. Success indicator: at least 3 groups identify at least one difference between the balloon and a magnet. Note any learners who say a balloon is 'magnetic' for later follow-up in Lesson 2. Collect the 'we wonder' statements as raw material for the DQB.
Driving Question Board (DQB) Creation	The class builds its first Driving Question Board on a chart or wall space. The central question is written at the top: 'Where do	Say: 'Good scientists keep track of what they wonder. Write at least two questions, one about what you saw and one about why it might	Driving Question Board (Asking Questions, NGSS SEP 1): learner-generated questions give ownership of the learning and	Check that every learner contributed at least two questions and that at least 70% are about cause or mechanism (why or how)

 VIDEO: Electric forces Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Charges and Fields <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	electric charges come from? Why does a rubbed balloon attract or repel other objects?' Each learner writes at least two questions on separate sticky notes, such as 'What is the invisible thing the balloon gives to my hair?', 'Why does the balloon stick longer in Nairobi than in Mombasa?', 'Is the shock from my jersey like lightning?' (Kipchoge), 'Why does my hair stand up on dry mornings?' (Wanjiru) and 'Why do two balloons push apart?'. In groups they cluster similar questions and give each cluster a heading, for example 'What is it?', 'Why does weather matter?', 'Why attract and why repel?', 'Is this lightning?', 'Where is it used?'. They place the clusters on the board and may add the KICD inquiry questions as the teacher reads them aloud. They agree which questions they hope to answer first.	happen.' Model a good question: 'Why does the balloon stick longer on some days?' and a weaker question: 'Is it good?'. Give 5 minutes for writing. WAIT TIME of 10 seconds before asking: 'Who will share a question that puzzles you most?' Cold-call 5 learners, including Amina, Wanjiru and Kipchoge, and place their notes on the board. Then ask groups to cluster the questions, giving 6 minutes. Walk around asking: 'What is the common idea in these questions?' Read aloud the seven KICD inquiry questions and ask: 'Which of your clusters match these?' Say: 'We will return to this board at the end of every lesson, moving questions we can answer and adding new ones.' Take a photo or label the board so it can be reused.	make the unit storyline visible. Clustering shows the links between questions and helps learners see that one big phenomenon needs several investigations, so each later lesson has a clear purpose.	rather than yes/no. Look for questions that link to weather (Amina), shocks and lightning (Kipchoge) and hair (Wanjiru). Success indicator: groups can explain the heading of their cluster in one sentence. Note common clusters to match them to Lessons 2 to 9.
Model Building Phase  VIDEO: Electric field direction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Balloons and Static Electricity Source: PhET Interactive Simulations (English)  Search ARES	Each learner draws an Initial Model on a sheet of paper (to be kept in a folder) showing the balloon, the hair, the wall and a paper piece before and after rubbing. They use arrows and labels to show what they think is happening, including anything they think moves, is made or is passed on, even if they cannot name it yet. They add a short caption: 'I think the balloon attracts the hair because ...'. A key box lists symbols they have used. In groups they compare their models, mark one similarity and one difference with a coloured pencil, and write one 'I am not sure about ...' statement. They finish by completing the Summary Table (What we observed / What we	Say: 'A model is a drawing that shows what you think is happening, including things we cannot see. There are no wrong models today. They are your starting point.' Show an example of a blank layout with four boxes (balloon, hair, wall, paper) but do not draw any particles. Give 10 minutes of silent drawing. Circulate and ask: 'What does your arrow show? Is something moving? What do you think is moving?' Do not correct. Then give 5 minutes for group comparison. WAIT TIME of 10 seconds before asking: 'What did someone else draw that you had not thought of?' Cold-call 4 learners. Say: 'Keep your model safe. In the next lesson we look inside materials to see what we	Initial Model Drawing (Developing and Using Models, NGSS SEP 2): putting ideas on paper before instruction exposes learners' current thinking and gives a baseline that can be revised in each lesson. Comparing models highlights different ideas, which produces genuine questions and shows the value of evidence in changing a model.	Check that each model includes all four objects (balloon, hair, wall, paper) and some form of cause (arrow, symbol or word) for attraction. Success indicators: at least 60% of learners label something invisible as moving or acting, and each learner writes one uncertainty statement. Note misconceptions such as 'the balloon is magnetic', 'the air pushes the paper' or 'glue-like stickiness' for targeted work in Lessons 2 and 3. Review Summary Tables for observations that are written as observations, not explanations.

for similar readings	learned / How we can explain it so far).	are made of, and you will revise it.' Close by returning to the DQB: 'Which question do you most hope we answer next?' Take 3 responses.		
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D. TEACHER REFLECTION

Which learner predictions and initial models revealed ideas about magnets, air or stickiness that need to be revisited in Lesson 2? Did all groups keep observation and explanation separate, and did they use the unrubbed balloon as a control? How many DQB questions were about cause rather than description? Did the dry-versus-humid contrast (Nairobi about 35%, Mombasa about 80%) prompt strong questions, and did Amina and Kipchoge feel their contributions were valued? What conditions (dry rooms, allergy-safe materials, balloon supply) need to be adjusted for the next lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	After 20 strokes on dry hair, the rubbed balloon made hair strands rise and follow it up to about 5 cm, stuck to the wall for about 3 minutes, and made 1 cm x 1 cm paper pieces jump up when held about 2 cm above them, with some falling off again. Two rubbed balloons on threads pushed apart. An unrubbed balloon did nothing. The effect is weak or absent on humid days (Mombasa, about 80%) and strong on dry days (Nairobi, about 35%).
What did I learn?	Rubbing makes the balloon able to pull on hair, wall and paper without touching them, and to push away another rubbed balloon. The effect is not the same as a magnet, it fades with time and it depends on the weather. We now have many questions on our DQB.
How does this explain the phenomenon?	So far we can only describe what happens: something invisible passes between the rubbed balloon and other objects, and rubbing and dry air seem to matter. Our initial models are hunches, and we do not yet know where the effect comes from or why the balloon sometimes attracts and sometimes repels.

LESSON 2 (80 minutes (2 x 40-minute periods)): What Are Electric Charges? Atoms, Protons, Electrons and the Rule of Attraction and Repulsion

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners explain what electric charges are by describing the atom (protons +, electrons -, neutrons 0), showing why neutral atoms have no overall charge, and testing with rubbed rods and strips the rule that like charges repel and unlike charges attract.
Knowledge	Learners describe an atom as a nucleus of protons ($+1.6 \times 10^{-19}$ C) and neutrons (0 C) surrounded by electrons (-1.6×10^{-19} C). A neutral atom has equal numbers of protons and electrons. Electric charge has symbol Q and unit coulomb (C). Like charges repel and unlike charges attract; this force is called the electrostatic force.
Skills	Making predictions from a rule; carrying out a simple investigation with rods and strips; recording and tabulating observations; using evidence to develop and revise a particle-level model (NGSS SEP: Developing and Using Models; Planning and Carrying Out Investigations; Constructing Explanations).
Attitudes	Curiosity about invisible particles; honesty in recording results even when they differ from predictions; cooperation when sharing apparatus; appreciation that scientists use evidence to infer what cannot be seen.
Key Inquiry Question	What are electric charges?
Purpose in Storyline	Lesson 1 left us with the observations that two rubbed balloons push apart and a rubbed balloon and hair pull together, and with many questions on the DQB. In this lesson learners find out what electric charge is and test the rule that like charges repel and unlike charges attract. This gives them the particle vocabulary (protons, electrons, neutrons) needed in Lesson 3, where they ask how rubbing gives the balloon and hair their charges.
Safety Notes	Use only light polythene strips and perspex rods; keep them away from faces. Teachers keep all materials dry (humidity makes the activity fail). Handle threads and stands carefully so rods do not fall. Do not use mains electricity. Keep the floor clear of dropped threads and paper pieces. Wash hands after handling cloths and materials.

B. LESSON OVERVIEW

Learners begin by recalling the Lesson 1 observation that two rubbed balloons on threads pushed apart while a rubbed balloon and hair pulled together, and predict what will happen with other rubbed objects. They then gather evidence in two ways: (1) a rods-and-strips investigation (polythene strip rubbed with wool/fur, perspex rod rubbed with silk) to test attraction and repulsion between pairs, and (2) a reading and card-sorting activity on the atom with charge values of the proton, electron and neutron. In the Explain phase they use both sets of evidence to answer: what are electric charges, why is a normal atom neutral, and what is the rule about like and unlike charges? Learners add questions to the DQB and draw a first particle-level model of the balloon, hair, wall and paper showing only what Lesson 2 supports: that matter contains positive and negative charges and that charges of the same sign repel while opposite signs attract. How the balloon and hair get their charges is left to Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Introduction to the atom Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Build an Atom Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners recall the Lesson 1 observation: two balloons, each rubbed on hair and hung on separate threads, pushed apart, while a rubbed balloon and hair pulled together. Individually they write for 3 minutes: 'If I rub a polythene strip with wool/fur and hang it on a thread, and bring a second identical rubbed strip near it, what will happen? What if I bring a rubbed perspex rod instead?' They write a prediction and a reason in their notebooks (a 'forces between charged objects' prediction table with three pairs: polythene-polythene, perspex-perspex, polythene-perspex). Learners also write one idea about what is inside a 'tiny piece' of balloon rubber that could make it push or pull.	Mr Mwangi displays the sentence from Lesson 1: 'Two rubbed balloons pushed apart; a rubbed balloon and hair pulled together.' Ask: "Last lesson we saw pushing and pulling with nothing touching. What do you think is inside the balloon and hair that could cause this?" Give 3 minutes of silent writing, then WAIT TIME of 10-15 seconds after each question before calling on anyone. Cold-call 4 learners (e.g. Otieno, Wanjiru, plus two others) to share predictions; ask each, "What makes you think that?" Do NOT confirm or correct. Record predictions for the three pairs on the board as tally marks (attract / repel / nothing). Ask Kipchoge's earlier question: "Is there a 'pushing' type and a 'pulling' type of invisible thing?" and leave it open.	Prediction with reasoning (Predict-Observe-Explain). Learners commit to a prediction and a reason before seeing data, which makes their current ideas visible and creates a 'need to know'. Linking to the Lesson 1 balloon observations connects the new investigation to the anchoring phenomenon.	Check notebooks: (1) learners have written a prediction for all three pairs; (2) at least one reason is given for each; (3) listen for ideas such as 'same things push, different things pull' or 'something tiny inside' versus 'magnetism'. Tally the class predictions and note the proportion expecting polythene-polythene to repel. Note learners who attribute the force to magnetism or electricity from a plug; these are targets for the Explain phase.
Observe Phase  VIDEO: Introduction to the atom - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Electric Charge and Electric Force (Flexbook) Source: CK-12  Search ARES for similar readings	Part A (about 25 minutes), groups of 4 with Mr Mwangi's dry materials: two polythene strips, two perspex rods, a wool/fur cloth, a silk cloth, thread and a stand or clamp. 1. Rub one polythene strip with wool/fur (20 strokes) and hang it from a thread. Rub a second polythene strip the same way and bring it near without touching. Record: attract, repel or no effect. 2. Repeat with two perspex rods each rubbed with silk (20 strokes). 3. Bring a silk-rubbed perspex rod near the hanging polythene strip. 4. As a control, bring an unrubbed polythene strip and an unrubbed perspex rod near the hanging strip. Learners fill the table: two polythene strips repel; two perspex rods repel; polythene and perspex attract; unrubbed objects show no effect. They note	Demonstrate hanging a strip and bringing the second near it, without giving results. Say: "Keep the strokes the same, 20 each, and keep everything dry so the effect shows clearly." Circulate with the prompt: "What exactly did the strip do when the second one came close? Describe it, do not explain it yet." Ask a group that gets no effect: "What could be affecting the rubbing or the dryness?" (Amina's humidity point from Lesson 1: damp materials let charge escape.) For Part B ask: "Which particle has no charge? How do you know from the table?" Wait 10-15 seconds before taking answers. Cold-call 3 groups to report their table results; ask others to show thumbs up if their results match. Chart the class results on the board. If a group's results conflict,	Data table and pattern-finding. Collecting the same measurements across groups lets learners see consistent patterns (same objects repel, different objects attract) rather than isolated events. The atom card sort gives a particle-level resource that makes charge a property of matter, preparing learners to use the pattern in the Explain phase.	Observable indicators: (1) learners fill in all rows of the rods-and-strips table including the unrubbed controls; (2) learners use the words attract and repel accurately; (3) the atom card sort shows 3 protons and 3 electrons for the neutral atom and learners calculate total charge 0 C; (4) learners can say the proton and electron charges have equal size (1.6×10^{-19} C) but opposite signs. Look for learners who think neutrons have charge or that the neutral atom has no charges inside at all.

	that the polythene strip rubbed with wool/fur is called negative and the perspex rod rubbed with silk is called positive (the teacher gives these labels; the reason is for Lesson 3). Part B (about 15 minutes): learners read the 'Inside the Atom' card set and complete the charge table: proton $+1.6 \times 10^{-19}$ C, electron -1.6×10^{-19} C, neutron 0 C. They sort cards to build a neutral atom (for example 3 protons, 4 neutrons, 3 electrons) and a second atom with 3 protons and 2 electrons, and calculate the total charge of each: 0 C for the first atom; $+1.6 \times 10^{-19}$ C for the second.	repeat the trial publicly. Do not use the phrase 'static electricity' as an explanation; use 'electric charge' and 'electrostatic force'.		
Explain Phase  VIDEO: 02 Electric Charge Source: Seavuria  Search ARES for similar videos  EXERCISE: Electric Charge and Electric Force Practice Source: CK-12  Search ARES for similar readings	In groups of 4 learners prepare a 'claim-evidence-reasoning' response to three questions on a poster: (1) What are electric charges and what carries them in an atom? (2) Why is a normal atom neutral if it contains charged particles? (3) What rule do the rods-and-strips results support, and how does it explain Lesson 1's two balloons pushing apart and balloon-hair pulling together? Learners use their tables: two polythene strips repel and two perspex rods repel (like charges repel); polythene and perspex attract (unlike charges attract). They write the rule 'Like charges repel, unlike charges attract' and state that the force is the electrostatic force. They apply it: two rubbed balloons both carry the same type of charge so they repel; a balloon and hair attract, so they carry opposite types. They compare their Predict-phase ideas with the evidence and write one sentence beginning 'I used to think ... now I think ...'. Learners also	Say: "Use evidence from our table, not just what you remember, to back up each claim." Give 5 minutes of silent individual planning, then 8 minutes in groups. Then run a gallery walk: each group posts a poster and 2 learners stay to explain while others rotate. After rotation, ask: "Which pair of results tells us that like charges repel?" WAIT TIME 10-15 seconds, then cold-call 3 learners. Ask: "Why do the balloons in Lesson 1 push apart if both were rubbed on hair?" Cold-call 2 more learners and probe: "What does that tell you about the charge on each balloon?" Press on the neutral atom: "If protons are positive and electrons are negative, why does an ordinary piece of rubber not attract everything?" Guide learners to equal numbers of protons and electrons. Correct, in Kenyan terms, the idea that charge is 'made': say "Charges are already in the atoms of the balloon, hair and cloth." Do not yet explain how	Claim-Evidence-Reasoning with gallery walk. Learners must connect each claim to specific rows in their data, which builds scientific argumentation. Peer explanation at the posters exposes gaps and lets learners refine their reasoning. Applying the rule back to the two balloons and balloon-hair observations from Lesson 1 ties the abstract atom idea to the anchoring phenomenon.	Look for: (1) posters state that protons are positive, electrons negative, neutrons uncharged, with the correct values; (2) the neutral-atom explanation mentions equal numbers of protons and electrons so charges cancel; (3) the rule is stated with matching evidence (polythene-polythene repel etc.); (4) learners apply the rule to the two-balloon observation. Use a 3-question exit check: 'What is the charge of an atom with 8 protons and 8 electrons?', 'Do two negative strips attract or repel?', 'What is the charge of a neutron?' Aim for 80% correct. Note learners who say a neutral atom has no charges at all, or who say unlike charges repel.

	answer: 'A neutral atom has 6 protons; how many electrons does it have, and what is the total charge?' (6 electrons; 0 C). Finally they note the unanswered issue: the particles that make charge are inside every atom, so why do ordinary objects not push or pull each other all the time? This is added to the DQB.	rubbing separates them (Lesson 3) or why neutral paper is attracted (Lesson 6). Record: 'Where does the unlike charge on the hair come from?' as a question for Lesson 3.		
Driving Question Board (DQB) Creation  VIDEO: Subatomic Particles (Proton, Neutron, Electron) - Example 1 Source: CK-12  Search ARES for similar videos  HTML: Build an Atom Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners return to the DQB from Lesson 1. In pairs they write new questions on sticky notes that came from today's evidence, such as 'What makes the polythene strip negative after rubbing?', 'Why did only electrons and not protons matter?', 'Why does the balloon also stick to a neutral wall if neutral atoms have no overall charge?', 'Is the charge on the balloon the same size as the charge on one electron?', 'Is the shock Kipchoge felt from a jersey like lightning?'. They place notes under the key question 'What are electric charges?' and move any Lesson 1 note that is now partly answered (for example 'why do two balloons push apart?') to a 'partly answered' section with a short evidence label such as 'like charges repel (rods and strips table)'. They add one 'we now know' card: 'Atoms contain + protons, - electrons and neutral neutrons; like charges repel and unlike charges attract.'	Say: "Look at the board from Lesson 1. Which questions can we now partly answer, and what new ones has today's evidence raised?" Give 3 minutes for pairs to write questions, then WAIT TIME 10-15 seconds before sharing. Cold-call 4 pairs to read and place a question; ask: "What evidence from today helps with this, or what do we still need?" Cluster questions by theme (how charge is gained, how charge moves, neutral objects attracted, shocks and lightning). Highlight Wanjiru's question about her hair after combing and point to the 'how are charges gained?' cluster for Lesson 3. Do not answer the new questions; say: "We will return to these as we collect more evidence."	Driving Question Board as a class record of knowledge and gaps. Learners see how evidence answers part of the driving question and how each answer produces new questions, creating a sense of purposeful progress across lessons.	Observe: (1) at least one new question per pair that links to today's evidence; (2) questions go beyond description (for example 'why' and 'how' questions); (3) learners can name the evidence that partly answers a moved question; (4) learners recognise which new questions need later lessons. Note pairs with only recall questions and prompt them with 'What could you ask about how the strip became charged?'
Model Building Phase  VIDEO: Electric forces Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners revise their Lesson 1 initial model, a labelled particle-level diagram of the balloon, hair, wall and paper. In their notebooks they draw a zoomed-in view of a small patch of balloon rubber and a hair strand, showing atoms as circles with a central nucleus labelled 'protons (+) and neutrons	Say: "Your model must show what we have evidence for today, and mark with a question mark what we do not know yet." Show a quick exemplar with an atom and legend on the board but leave the balloon and hair incomplete. Give 10 minutes of silent drawing, then 5 minutes of pair sharing. WAIT	Iterative model revision. Learners return to their original model, add the evidence-based ideas from this lesson and mark uncertainties. This makes the growth in understanding visible and models how scientists refine explanations as new evidence appears.	Indicators: (1) atoms drawn with protons and neutrons in the nucleus and electrons outside; (2) equal numbers of + and - in neutral atoms; (3) arrows or labels for repulsion between like charges and attraction between unlike charges; (4) the model avoids claiming charge moves or is

 HTML: Electric Charge and Electric Force (Flexbook) Source: CK-12  Search ARES for similar readings	(0)' and electrons (-) around it, with equal numbers of + and - so each atom is neutral. They then add arrows and labels for the Lesson 1 observations they can now support: two balloons pushing apart (like charges repel, arrow labelled 'electrostatic force') and balloon and hair pulling together (unlike charges attract). They mark balloon with '?' for 'how did it get its charge?' and write 'Model revision 2: what changed and why?' listing evidence (rods and strips table; charge values of proton $+1.6 \times 10^{-19}$ C, electron -1.6×10^{-19} C, neutron 0 C). They add a key and a note of what the model cannot yet explain: how the balloon and hair got their charges, why neutral paper and wall are attracted.	TIME 10-15 seconds before asking: "What did you add or change from your first model, and what evidence made you do that?" Cold-call 3 learners to explain their change. Prompt: "Does your model show any particle moving yet?" (not yet; that is Lesson 3). Remind learners to keep their model for Lesson 3. Collect notebooks for quick review.		created; (5) a short 'what changed and why' statement referring to evidence. Common gaps: placing electrons in the nucleus, or showing unequal numbers of + and - in neutral objects.
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D. TEACHER REFLECTION

After the lesson reflect: Did most learners get consistent repel/attract results in the rods-and-strips activity? If some groups saw no effect, was dryness or rubbing the cause (Amina's humidity point)? Could learners use the words proton, electron and neutron correctly and say why a neutral atom has equal numbers of protons and electrons? Did any learners think charge is created or that positive charge moves in solids? (Note these for Lesson 3, where electron transfer is introduced.) Which DQB questions show curiosity about how the balloon and hair got their charges? Did learners rely on evidence from the table rather than memory in their claims? What would you change in pacing for the Observe phase?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Two polythene strips rubbed with wool/fur repel; two perspex rods rubbed with silk repel; a polythene strip and a perspex rod attract; unrubbed strips and rods do nothing. Proton charge is $+1.6 \times 10^{-19}$ C, electron charge is -1.6×10^{-19} C and neutron charge is 0 C.
What did I learn?	Atoms contain positive protons and neutral neutrons in the nucleus with negative electrons around it. A neutral atom has equal numbers of protons and electrons, so the total charge is 0 C. Like charges repel and unlike charges attract through the electrostatic force.
How does this explain the phenomenon?	The two rubbed balloons in Lesson 1 pushed apart because they carry the same type of charge, while the balloon and hair attracted because they carry opposite types. We still need to find out how rubbing gives objects their charge and why neutral objects such as paper and the wall are also attracted.

LESSON 3 (80 minutes (2 x 40-minute periods)): Charging by Friction: Who Wins the Electrons?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners rub pairs of materials together, use the repulsion test to find the sign of charge on each, and use the triboelectric table to explain that friction transfers electrons so the balloon becomes negative and the hair positive.
Knowledge	Friction transfers electrons from one material to another. Protons stay in the nucleus. The sign of charge depends on the pair of materials (triboelectric order: dry human hair (+), glass, perspex, wool/fur, silk, polythene, ebonite, latex balloon (-)). The material higher in the list becomes positive when rubbed with one lower in the list. Charge is not created: the two materials gain equal and opposite charge and total charge is conserved. A rubbed balloon gains about 1 nC, about 6×10^9 extra electrons, out of roughly 10^{23} atoms.
Skills	Developing and using models (particle-level diagrams of electron transfer); planning and carrying out investigations (rubbing pairs, repulsion test); analysing and interpreting data (ranking materials, finding patterns in a table); using mathematics (charge in nC to number of electrons); constructing explanations.
Attitudes	Curiosity about invisible forces; honesty in recording results even when they differ from predictions; cooperation in groups; respect for evidence; care when handling materials. Learners value careful observation and using evidence to revise their models.
Key Inquiry Question	Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? (Focus: What happens to the electrons when two materials are rubbed together, and which material ends up positive or negative?)
Purpose in Storyline	Lesson 1 gave us the puzzling observations: hair rises, the balloon sticks to the wall and paper jumps. Lesson 2 gave us the tools: protons (+), electrons (-), neutrons (0), neutral atoms, and the rule that like charges repel and unlike charges attract. Lesson 3 answers the first part of the driving question: where charges come from. Learners find that charge is not created but transferred as electrons by friction, and they assign the balloon as negative and the hair as positive. This provides the transfer mechanism that Lessons 4, 5 and 6 will build on (conductors, contact, and induction explaining the neutral wall and paper).
Safety Notes	Use only dry, clean materials and low-energy rubbing; no mains electricity or high-voltage equipment. Keep balloons away from eyes and do not put them over the mouth. Check for latex allergy before the balloon activity and offer a polythene strip instead. Hang threads from desk edges or retort stands, not from the ceiling. Keep the stations tidy so learners do not trip over threads. A mild shock from a jersey is harmless, but learners should not deliberately try to shock others.

B. LESSON OVERVIEW

Mr Mwangi opens with the hair and balloon from Lesson 1 and asks: what is the balloon taking from, or giving to, the hair? Learners predict, then rotate through stations rubbing polythene, perspex, ebonite, glass, silk, fur and a latex balloon in pairs. Each rubbed object is hung on a thread and tested with the Lesson 2 repulsion test against a known negative polythene strip (rubbed with wool/fur) and a known positive perspex rod (rubbed with silk). From the class data they build a triboelectric order and compare it with the official table. In the Explain phase they draw particle-level models of electron transfer for the balloon and hair, check that total charge is conserved, and calculate that 1 nC is about 6×10^9 electrons, a tiny fraction of the roughly 10^{23} atoms in a balloon. They add new questions to the DQB (why does the balloon stick to the neutral wall? Why does it fail in Mombasa's humidity?) and revise their initial models from Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Conservation of charge Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners recall the balloon and hair from Lesson 1 (20 strokes, hair rises up to 5 cm). On a prediction card they answer individually: (1) After rubbing, is the balloon charged? Is the hair? (2) What do you think moved between them? (3) If the balloon is negative, what sign is the hair? Wanjiru says her hair rises after combing on dry Nairobi mornings; learners predict whether the comb and her hair have the same or opposite charge. Then, in pairs, they predict which material, polythene or perspex, will become positive when rubbed with silk, and explain why. They record predictions in their notebooks with a sketch of their current idea.	Hold up a latex balloon and a lock of hair. Say: "Last week the balloon made hair rise. Today we ask a more precise question: what exactly happened to the hair and to the balloon when I rubbed them?" Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds after asking: "What do you think moved from one to the other, if anything?" Cold-call 4 learners, including at least one who has not spoken yet. Record every prediction on the board, without evaluating, in three columns: 'Charge is made', 'Something moves', 'Not sure'. Ask: "Wanjiru, does your comb make charge, or take it from your hair? How could we test that?" Do not reveal electrons yet; use Lesson 2 vocabulary (protons +, electrons -, neutrons 0) and ask learners which particles could possibly move in a solid.	Eliciting prior ideas through Predict-Explain: learners commit to a prediction and a reason before seeing data. Making ideas public surfaces the common misconception that rubbing creates charge, so the later evidence has something to confront and learners can see how their thinking changes.	Look for: (a) learners who say 'rubbing creates charge' versus 'charge moves'; (b) whether they apply the Lesson 2 rule that like charges repel and unlike charges attract; (c) whether they name electrons as the particles most likely to move. Tally the three columns of the board; target at least 70% of learners giving a reason, not just a guess. Note learners who say positive charge moves, to address in the Explain phase.
Observe Phase  VIDEO: Conservation of charge Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Charges and Fields <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES	In groups of 4, learners rotate through 3 stations (about 8 minutes each). Station A (Known signs): rub a polythene strip with wool/fur and a perspex rod with silk, hang each on a thread, and confirm that two polythene strips repel, two perspex rods repel and polythene and perspex attract (Lesson 2 results), giving a negative reference (polythene) and positive reference (perspex). Station B (Pairs): rub each pair with 20 strokes: glass with silk, ebonite with fur, perspex with wool/fur, polythene with silk, and a balloon with dry hair. Test each rubbed object against both references. Only repulsion counts as proof of sign; attraction alone	Before the stations, say: "Attraction is not enough. A neutral object is also attracted to a charged one. Which result proves the sign?" WAIT TIME of 10-15 seconds, then cold-call 2 learners to say 'repulsion'. While circulating ask: "Which one repelled the polythene? What sign does that make it?" and "Did the cloth gain the same sign as the rod or the opposite?" Prompt: "Why did you test an unrubbed object?" Remind groups of 20 strokes each and a dry surface; if a group gets weak results, ask: "What could be different about the air or the materials?" (link to Amina's Mombasa humidity). Keep reference rods away from other	Data pattern-finding with a class results table: learners compare their observed order with the table and look for the pattern that the material higher in the list becomes positive and the one lower becomes negative. Using a controlled test (unrubbed control, two references) builds the reasoning that repulsion is the reliable sign test.	Observable indicators: learners use repulsion, not attraction, as the evidence for sign; the table has a sign for every material with a stated reason; groups find that the rod and cloth have opposite signs; the class order roughly matches the triboelectric table. Check that the balloon is recorded as negative and hair as positive. Ask each group to show one row of evidence (e.g. 'perspex repelled perspex, so it is positive'). Note any group that reports attraction as proof of sign for re-teaching.

for similar readings	could be a neutral object. Record in a table: material, repelled by polythene or perspex, sign. Station C (Both partners): test that the cloth also becomes charged, with the opposite sign to the rod, by hanging the cloth or hair and testing it. Learners also test an unrubbed object as a control, which should show no repulsion. Finally, the class pools results and ranks the materials into an order from most positive to most negative.	materials so they stay charged. After the stations, collect results on a class chart and ask 3 groups to report, cold-calling a different learner in each. Ask: "Where does the balloon-hair pair fit in the order? Which one is higher?" Show the triboelectric table only after learners have their own order: dry human hair (+), glass, perspex, wool/fur, silk, polythene, ebonite, latex balloon (-).		
Explain Phase VIDEO: Conservation of charge Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	Learners use their data to explain why the balloon becomes negative and the hair positive. Step 1: they draw a particle-level diagram of the hair and balloon before rubbing (both neutral: equal + and - symbols). Step 2: they draw after rubbing: the balloon has extra electrons (-), the hair has a deficit (+), with an arrow showing electrons moving from hair to balloon. They explain why the hair strands, now both positive, repel each other and rise, and why they are attracted to the negative balloon (Lesson 1 and 2 rule). Step 3: charge conservation: they count symbols to show the total is the same before and after (e.g. 6 electrons moved: balloon -6, hair +6, total 0). Step 4: a calculation: a balloon gains about 1 nC; number of electrons = $1 \times 10^{-9} / 1.6 \times 10^{-19} = 6.25 \times 10^9$, about 6×10^9, compared to about 10^{23} atoms in the balloon, so only a tiny fraction of electrons move. Kipchoge asks if the shock from a jersey is like lightning, and learners explain that the jersey and body exchange electrons by friction and the built-up charge discharges as a small spark on a	Say: "Use evidence from today's table. Why is the balloon negative and the hair positive?" WAIT TIME of 10-15 seconds, then pair-share for 2 minutes. Cold-call 3 pairs. Press for precision: "Which particles moved? Did protons move? Why not?" (they are in the nucleus). Correct any statement of 'positive charge moved' with: "In a solid, only electrons move. Positive means a deficit of electrons." Ask: "Latex holds electrons more strongly than hair; where is that on our table?" Model the calculation on the board, then ask learners to do it for 2 nC. Ask: "If the balloon gains 6×10^9 electrons, how many does the hair lose?" (the same number, so charge is conserved). Ask: "Is charge created or transferred?" Do not explain why the balloon sticks to the wall or the paper jumps; say: "That is a new question for the DQB. We will need induction later." Do not discuss conductors or contact charging, which are for Lessons 4 and 5. Invite Kipchoge's question into the discussion and say lightning will be handled in Lesson 7.	Modelling with particle-level diagrams and a claim-evidence-reasoning (CER) frame: Claim (balloon is negative, hair positive), Evidence (repulsion tests, table), Reasoning (electron transfer, conservation of charge). Drawing before and after states makes the invisible transfer concrete, and the calculation shows the scale of the transfer.	Look for: diagrams where electrons, not protons, move; equal and opposite charge on balloon and hair; a total of zero before and after; correct calculation of 6×10^9 electrons for 1 nC and a correct scaled answer for 2 nC (about 1.25×10^{10}); correct use of 'transfer' not 'create'; the CER statement that cites repulsion evidence. Use a quick exit check: 'Hair is positive because it has... (a deficit of electrons).' Aim for at least 80% correct. Note learners who say positive charges moved.

	dry Nairobi day (35% humidity).			
Driving Question Board (DQB) Creation  VIDEO: Conservation of charge Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Charges and Fields <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners return to the DQB from Lesson 1. In groups they write on sticky notes: (1) a question now answered (e.g. 'Does rubbing make charge?'), with the evidence from today, and move it to the 'Answered' section; (2) new questions raised by today's work (e.g. 'Why does the balloon stick to the neutral wall if the wall has no charge?', 'Why did Amina's balloon not work in Mombasa's rainy season?', 'Why do some materials, like metal, not hold charge when rubbed?', 'Is the jersey shock like lightning?'). They sort questions by lesson ('Where do charges come from?' is now mostly answered; 'How do we detect charge?' and 'Why does charge stay on the balloon?' remain open).	Say: "Look at the DQB. Which of our questions can we now answer with today's evidence?" WAIT TIME of 10-15 seconds. Cold-call 3 groups to move a note and give the evidence. Then ask: "What new questions do you have?" Cold-call 3 more learners. Ask: "Otieno, you saw the balloon stick to the wall; does today's rubbing explain the wall?" and guide them to see that the wall was not rubbed and so needs another explanation (a link to later lessons). Do not answer open questions; label them with the likely lesson (conductors in Lesson 4, contact in Lesson 5, induction and the wall in Lesson 6, points and lightning in Lesson 7). Photograph the DQB.	Driving Question Board as a public record of understanding: moving questions to 'answered' with cited evidence shows progress, and the new questions create a need to know that motivates the following lessons.	Indicators: at least one question per group is moved to 'Answered' with a piece of evidence; new questions are specific and linked to the phenomenon (wall, paper, humidity, shock); at least two groups identify the wall question as unexplained by friction. Check that learners do not claim 'answered' without evidence.
Model Building Phase  VIDEO: Conservation of charge Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners revise the model they began in Lesson 1 and updated in Lesson 2: a labelled particle-level diagram of the balloon, hair, wall and paper. In this lesson they add: the balloon with excess electrons (-), the hair with a deficit (+) and an arrow for electrons moving from hair to balloon, the label '~1 nC = ~6 x 10⁹ electrons', and a note that total charge is conserved. They draw the hair strands repelling each other and being attracted to the balloon. For the wall and paper, they draw them neutral, with a question mark and the note 'Why attracted? (Lesson 6)'. They write a 2-3 sentence caption using 'transfer of electrons' and finish the summary table.	Say: "Revise your model with today's evidence. Show the electrons moving. Label the signs." Circulate and ask: "Where did the electrons come from? Where did they go? Is the total charge the same?" WAIT TIME of 10-15 seconds before taking answers. Cold-call 3 learners to explain one part of their model. Say: "Keep the wall and paper neutral for now. Mark what you cannot yet explain." Insist on correct vocabulary: 'electrons are transferred', not 'charge is made'. Collect models for review. Preview Lesson 4: "Why does the charge stay on the balloon? Why does it not stay on a metal spoon? We will test that next time."	Iterative model revision with gaps made explicit: learners add new evidence and mark the parts the model cannot yet explain (wall, paper, humidity), so models are treated as tools that improve as evidence grows.	Check models for: electron arrow from hair to balloon; balloon negative and hair positive; equal numbers of transferred charges; the 1 nC and 6 x 10⁹ label; hair strands repelling; wall and paper neutral and flagged as unexplained; no protons shown moving. Success: at least 80% of models include all of these. Use the caption for individual feedback.

D. TEACHER REFLECTION

After the lesson ask: Did learners use repulsion, not attraction, as evidence of sign? Did anyone still say charge was 'made' or that positive charges moved? Which groups had weak results, and was humidity or damp materials the cause? Did the table pattern emerge from their data before I showed the official triboelectric table? Did the calculation (1 nC to 6×10^9 electrons out of 10^{23} atoms) help them see the scale? Were quieter learners heard during the cold-calls? Did I resist explaining the wall and paper, leaving that to Lesson 6? What should I change in the Lesson 4 conductor investigation so that learners connect free electrons to why the balloon holds charge?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We rubbed pairs of materials (glass with silk, ebonite with fur, perspex with wool/fur, polythene with silk, balloon with hair) and tested them against a negative polythene strip and a positive perspex rod. Rod and cloth had opposite signs, shown by repulsion. The balloon was negative and the hair positive. Unrubbed objects showed no effect.
What did I learn?	Friction transfers electrons from one material to the other. Protons stay in the nucleus. The sign depends on the material pair, following the triboelectric order: dry human hair (+), glass, perspex, wool/fur, silk, polythene, ebonite, latex balloon (-). Charge is conserved: the balloon gains about 1 nC (6×10^9 electrons) and the hair loses the same amount, out of about 10^{23} atoms.
How does this explain the phenomenon?	When I rub the balloon on my hair, electrons transfer from the hair to the balloon because latex holds electrons more strongly. The balloon becomes negative and the hair positive. The hair strands, all positive, repel each other and rise, and they are attracted to the negative balloon. Charge is not created. Still to explain: why the balloon sticks to the neutral wall and attracts paper, and why humidity makes it fail.

LESSON 4 (80 minutes (2 x 40-minute periods)): Conductors and Insulators: Why Does the Balloon Keep Its Charge While a Metal Spoon Does Not?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners test materials in a simple circuit to separate conductors from insulators, and use free (delocalised) electrons to explain why the rubbed latex balloon keeps its charge and sticks to the wall.
Knowledge	Conductors (copper, aluminium, steel spoon, graphite) have free (delocalised) electrons that move through the material. Insulators (plastic, dry wood, rubber, glass, dry cotton) hold their electrons tightly bound. Only electrons move in solids; protons stay in the nucleus.
Skills	Planning and carrying out an investigation (NGSS SEP 3): build a circuit with a 3 V cell pair and torch bulb, test 9 materials and record bulb lit or not lit. Developing and using models (SEP 2): draw particle-level diagrams of free versus bound electrons. Constructing explanations (SEP 6).
Attitudes	Curiosity about invisible particles, honesty in recording results even when they surprise, safe and careful handling of circuits, and respect for each other's ideas during discussion.
Key Inquiry Question	Why do some materials conduct electricity while others don't?
Purpose in Storyline	Lessons 2 and 3 showed that electrons are the movers and that friction transfers them to the balloon. Learners now ask why the balloon's charge stays on it instead of spreading away. This lesson gives the idea of free versus bound electrons that is needed for contact charging (Lesson 5), induction and earthing (Lesson 6) and distribution of charge (Lesson 7).
Safety Notes	Use only the 3 V supply (two 1.5 V cells in series). Never connect to mains sockets. Do not short-circuit the cells with bare wire for more than a few seconds because wire and cells heat up. Keep spoons and wires away from the face. Wash hands after handling graphite and metals. Keep all materials dry; teachers always use dry conditions.

B. LESSON OVERVIEW

Mr Mwangi reopens the balloon phenomenon: the rubbed balloon stays charged and sticks to the wall for about 3 minutes on a dry Nairobi day, but a metal spoon rubbed with wool never seems to hold charge. Learners predict which of 9 materials will light a bulb, build a circuit with a 3 V cell pair and torch bulb, and test copper wire, aluminium foil, a steel spoon, graphite, a plastic ruler, dry wood, rubber, a glass rod and dry cotton cloth. They sort results, explain them with free versus tightly bound electrons, and revise the DQB and their particle-level model to show why charge stays put on the insulating latex balloon. Contact charging, induction and earthing are left for Lessons 5 and 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Conductors and	Mr Mwangi holds up a rubbed balloon stuck to the wall and recalls Otieno's observation from	Say: 'Last lesson we learned that friction transfers electrons to the balloon. Today we ask: what lets	Predict-Observe-Explain (POE) with a class tally: committing to written predictions before evidence	Check that 100% of learners have written 9 predictions with at least 3 reasons. Look for reasons using

insulators Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Circuit Construction Kit: DC Source: PhET Interactive Simulations (English) Search ARES for similar readings	Kisumu that it stays for about 3 minutes. He then holds a steel spoon and asks, 'If electrons moved onto the balloon from the hair, why do they stay put on the balloon?' Learners individually write a prediction for 9 materials (copper wire, aluminium foil, steel spoon, graphite pencil lead, plastic ruler, dry wood, rubber, glass rod, dry cotton cloth): will the bulb light in a circuit with a 3 V cell pair, Yes or No? They give a one-line reason, then share in pairs and mark their predictions on a class tally on the board.	electrons move through some materials but not others?' Ask: 'Which of these 9 materials do you think will let electricity pass? Write Yes or No and a reason.' WAIT TIME: 10-15 seconds of silent writing before any talk. Give pairs 2 minutes. Cold-call 4 learners (e.g. Amina, Kipchoge, Wanjiru, Otieno) for one prediction each and the reason, and record on the board without judging. Ask: 'Who changed their mind after hearing a partner?' Do not reveal answers.	creates cognitive conflict when results surprise learners, and the recorded reasons expose preconceptions (for example, that all metals or all solids conduct, or that wood conducts) that the evidence can then test.	ideas from Lessons 2 and 3 (electrons move, protons stay). Note the number of learners predicting that plastic or wood conducts, and the number who mention 'metal' only as a reason. Record the tally for comparison after the test.
Observe Phase  VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) Source: CK-12 Search ARES for similar readings	In groups of 4, learners build a simple circuit with two 1.5 V cells in series (3 V), a torch bulb and connecting wires, leaving a gap. They touch the gap ends to each material in turn: copper wire, aluminium foil, metal (steel) spoon, graphite (pencil lead), plastic ruler, dry wood, rubber, glass rod, dry cotton cloth. Results: the bulb lights with copper wire, aluminium foil, steel spoon and graphite; it does not light with plastic ruler, dry wood, rubber, glass rod or dry cotton cloth. They record bulb Lit or Not lit in a table, compare with their predictions, and mark each as conductor or insulator. They then link back to the phenomenon: the balloon is latex rubber, which does not light the bulb.	Demonstrate one safe circuit with copper first, then circulate. Say: 'Make sure the material completes the gap, with both ends firmly touching.' Ask: 'What would a bulb that does not light tell you? Is the circuit broken, or is the material not letting electrons pass?' Tell groups to first test the bulb by touching the two wire ends together. WAIT TIME: 10-15 seconds before groups record each result to allow them to check contacts. Prompt: 'Which result surprised your group? Is graphite a metal?' Ask each group to retest any 'odd' result once. Remind: 'Keep materials dry. A damp cloth could give a different result.' Cold-call 3 groups to report the two materials they tested last.	Structured data table with pattern-finding (SEP 3 and SEP 4): recording lit or not lit for all 9 materials makes a consistent pattern visible, and retesting odd results teaches reliability of evidence. Graphite is a key case because it is not a metal, so a 'metals conduct' rule is not enough and learners need a mechanism.	Observe whether groups complete the circuit correctly and test the bulb first. Check that all 9 results are recorded with 4 materials lit and 5 not lit. Look for correct use of the words conductor and insulator. Note groups that retest an unexpected result rather than changing it to match their prediction.
Explain Phase  VIDEO: Metallic bonds <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES	Learners sort the 9 materials into conductors (copper, aluminium, steel spoon, graphite) and insulators (plastic ruler, dry wood, rubber, glass, dry cotton). Using a model of a metal as a lattice of fixed positive ions surrounded by free (delocalised) electrons, they draw how electrons drift through a	Say: 'Only electrons move in solids; protons stay in the nucleus.' Ask: 'What is different about the electrons in copper and in the plastic ruler?' WAIT TIME: 10-15 seconds. Cold-call 3 learners to describe free electrons in their own words, and 2 more to challenge or add. Ask: 'Graphite is not a metal.	Modelling with particle-level diagrams and Claim-Evidence-Reasoning (SEP 2 and 6): claim (steel spoon is a conductor), evidence (bulb lit), reasoning (free electrons move through the material). Drawing bound versus free electrons links the observable bulb result to the invisible particle	Check each pair's explanation for three indicators: (1) says electrons, not protons, move; (2) says conductors have free or delocalised electrons and insulators have tightly bound electrons; (3) links the latex balloon being an insulator to the charge staying on it. Ask learners

for similar videos EXERCISE: Electric Conductors and Insulators Practice Source: CK-12 Search ARES for similar readings	conductor from the negative terminal of the cell, and how in an insulator electrons stay tightly bound to their atoms. In pairs they write a 3-sentence explanation: why does the bulb light with the steel spoon but not with the plastic ruler? Then they apply it to the balloon: because latex is an insulator, the extra electrons gained from the hair (about 6×10^9 for 1 nC) cannot move through the balloon and stay where they were deposited, so the balloon keeps its charge and stays stuck to the wall for about 3 minutes on a dry day.	What does its bulb result tell us about its electrons?' Draw the two particle diagrams on the board while learners draw theirs. Ask: 'Why do we say a conductor has free electrons, not free protons?' Then ask: 'Why does the extra charge on the rubbed balloon stay on the balloon?' Do not teach charging by contact or earthing; say: 'Lesson 5 will ask what happens when a charged object touches a conductor.'	explanation, and applying it to the balloon connects it to the anchoring phenomenon.	to hold up fingers (1-3) for how many indicators they included. Flag learners who say charge is created or that positive charge moves in solids.
Driving Question Board (DQB) Creation VIDEO: Current direction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Circuit Construction Kit: DC Source: PhET Interactive Simulations (English) Search ARES for similar readings	Learners return to the DQB from Lesson 1. They add sticky notes with (a) answers they now have, such as 'Rubber balloons are insulators, so the electrons stay on them' and (b) new questions such as 'If metals let electrons move, can we charge a spoon?' (Kipchoge), 'Why did my hair stand up less in humid Mombasa?' (Amina), and 'Would a metal balloon hold charge?' (Otieno). They group notes under KICD questions 2 (What are electric charges?) and 3 (Why do some materials conduct and others don't?), and mark questions to be answered in Lessons 5 to 8.	Say: 'Choose one question you can now partly answer and one new question we cannot answer yet.' WAIT TIME: 10-15 seconds for writing. Cold-call 4 learners to read a note and place it on the board. Ask: 'Which earlier question does the conductor-insulator idea help with?' Ask the class to tell you whether each new question is about contact, induction, distribution or detecting charge, and tag it with the matching lesson number. Keep Amina's humidity question open and ask: 'What might water vapour do to charge on an insulator?'	Driving Question Board with question-clustering: sorting notes shows learners how the evidence builds toward the driving question and how their own questions shape upcoming lessons, which keeps learners as owners of the inquiry.	Count the notes: aim for at least 1 new answer and 1 new question per learner. Check that answers use free or bound electrons correctly. Look for questions that point forward (contact, induction, earthing, spreading of charge) as evidence of curiosity and understanding of the sequence.
Model Building Phase VIDEO: 2015 AP Chemistry free	Learners revise their labelled particle-level model of the balloon, hair, wall and paper. They add: (1) the balloon as an insulating latex surface where the extra electrons (about 6×10^9 for 1 nC out of	Say: 'Use a different colour to add today's idea to your Lesson 3 model. Show only what the evidence supports.' Ask: 'Where on your model are the free electrons? Where are the bound	Iterative model revision (SEP 2): adding a new layer of explanation in a different colour makes the growth of understanding visible, and labelling the gaps with question marks identifies what	Check each model for: electrons shown on the balloon, a label saying latex is an insulator, protons not moving, and a clearly marked unexplained gap. Look for correct signs (balloon negative,

response 2d and e <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Electric Conductors and Insulators (Flexbook) Source: CK-12 Search ARES for similar readings	roughly 10^{23} atoms) stay in place; (2) hair as positive; (3) a note that the wall and paper have electrons that are tightly bound but can shift slightly, which they will test in Lesson 6. They add an arrow labelled 'a metal spoon would let electrons move' and a caption 'Latex is an insulator: electrons stay where friction put them.' They write one sentence explaining why the balloon sticks for about 3 minutes on a dry day.	electrons?' WAIT TIME: 10-15 seconds before pairs compare. Cold-call 3 learners to explain their revised diagram. Say: 'We have not yet explained why the wall attracts the paper; leave that gap labelled with a question mark.' Ask: 'What can your model not yet explain?' Collect models for review.	later lessons must explain, so the model becomes a tool for prediction and not just illustration.	hair positive). Note any learner who shows positive charges moving in the solid and revisit it in a short conference.
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D. TEACHER REFLECTION

Which learners explained conductors using free electrons rather than just saying 'metals conduct'? Did graphite help to move learners beyond the rule 'metals conduct'? Did anyone say that an insulator cannot be charged, and how did you address it with the balloon? Which predictions changed most after the circuit test? Which DQB questions should be prioritised in Lesson 5 and Lesson 6? Were all tests done in dry conditions, and did any result differ because of damp materials?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	The bulb lit with copper wire, aluminium foil, a steel spoon and graphite. It did not light with a plastic ruler, dry wood, rubber, a glass rod or dry cotton cloth in the 3 V circuit.
What did I learn?	Conductors have free (delocalised) electrons that can move through the material; insulators have tightly bound electrons. Only electrons move in solids. The latex balloon is an insulator, so the electrons it gains from the hair stay on it.
How does this explain the phenomenon?	The balloon is negative and holds its extra electrons (about 6×10^9 for 1 nC) because latex is an insulator, so the charge cannot move away and the balloon stays charged and sticks for about 3 minutes on a dry day.

LESSON 5 (40 minutes): Charging by Contact: Why Do Paper Pieces Touch the Balloon and Then Fall Off?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate what happens when a conductor is touched by a charged object, so that they can explain the sign of charge the conductor gains using electron movement. This advances the driving question by explaining why paper pieces that touch the balloon later fall off.
Knowledge	A conductor touched by a charged object gains the SAME sign of charge by transfer of electrons. A negative polythene strip touched to the electroscope cap leaves the leaf at about 30 degrees after the strip is removed. Electrons move because conductors have free electrons (Lesson 4). Charge is transferred, not created (Lesson 3).
Skills	Developing and using models (particle-level diagrams of electron transfer); planning and carrying out investigations (electroscope observations); analysing and interpreting data (leaf divergence in degrees before, during and after contact); constructing explanations using evidence; communicating in pairs and groups.
Attitudes	Curiosity about invisible forces; careful, honest observation and recording; respect for others' ideas during discussion; perseverance when a result is unexpected; safe handling of equipment.
Key Inquiry Question	How can we charge conductors? (links to the driving question: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects?)
Purpose in Storyline	Lessons 3 and 4 showed that friction transfers electrons and that conductors have free electrons. Lesson 5 asks what happens when a charged object TOUCHES a conductor. It explains why paper pieces that touch the balloon sometimes fall off, and gives the rule that contact gives the SAME sign. Lesson 6 will contrast this with induction, which gives the OPPOSITE sign.
Safety Notes	Use only dry, low-charge equipment (polythene strip rubbed with wool, 20 strokes). The charges are very small and harmless. Handle the gold-leaf electroscope gently because the leaf is fragile, and do not breathe or blow on it. Keep the equipment dry, because humidity makes the charge leak away. Tidy away torn paper and cloths after the lesson.

B. LESSON OVERVIEW

Mr Mwangi reminds the class of the puzzle from Lesson 1: some paper pieces jumped up to the rubbed balloon, touched it, and then fell off. Learners first predict what will happen to a neutral electroscope when a negative polythene strip touches its cap. They then observe the leaf rise to about 30 degrees and stay there after the strip is removed, and they watch paper pieces behave in the same way at the balloon. In the Explain phase they use free electrons (Lesson 4) and electron transfer (Lesson 3) to explain that the cap, stem and leaf gained extra electrons and so became negative, the same sign as the strip. They connect this to the paper pieces: after touching the negative balloon, a piece gains the same sign, so it is repelled and falls off. The class updates the Driving Question Board and revises the particle-level model of the balloon, hair, wall and paper.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the balloon on the	Say: "Remember the paper pieces	Predict-Observe-Explain	Look for learners who: (1) sketch

 VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	wall and the paper pieces that jumped to the balloon and then fell off (Lesson 1). Mr Mwangi shows a neutral gold-leaf electroscope with the leaf hanging at 0 degrees and a rubbed polythene strip (negative, from Lesson 2). Individually, in their notebooks, learners predict what will happen to the leaf if the strip is touched to the cap and then removed. They sketch the leaf before, during and after contact and write one sentence of reasoning. They then compare predictions with a partner. Wanjiru predicts that the leaf will only rise while the strip is there. Otieno predicts that it will stay up.	that jumped up to the balloon and then fell off again. Today we find out why." Hold up the neutral electroscope and point to the leaf at 0 degrees. Ask: "If I touch this negative strip to the cap and then take it away, what will the leaf do? Sketch it before, during and after." Give 2 minutes of silent individual writing. WAIT TIME: wait 10-15 seconds after asking for reasoning before taking any answers. Then cold-call 3 learners (for example Wanjiru, Otieno and Amina) to share a prediction and a reason, and record the predictions in 3 columns on the board: "falls back to 0 degrees", "stays up", "goes higher". Do not say which is correct.	(prediction stage): committing to a written prediction with a reason makes learners' current ideas about charge visible. It sets up cognitive conflict if the leaf does not fall back after the strip is removed, so learners need to revise their thinking in the later phases.	three separate leaf positions (before, during, after); (2) give a reason that mentions electrons, charge or the strip, not only "it just moves"; (3) use the term 'leaf divergence' or 'leaf rises' correctly. Note on a class list who predicts that the leaf returns to 0 degrees after removal, because this shows the idea that charge only stays while the charged object is near. Count how many learners mention electrons in their reasoning (aim for at least 5 of 10 in a quick scan).
Observe Phase  VIDEO: Conductors and insulators Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	In groups of 4, with a gold-leaf electroscope, a polythene strip and a wool or fur cloth per group (Mr Mwangi demonstrates first), learners: (1) check that the leaf is at 0 degrees; (2) rub the polythene strip with wool for 20 strokes (negative, from Lesson 2); (3) bring the strip near the cap without touching: the leaf rises to about 30 degrees and falls back when the strip is taken away; (4) discharge the electroscope by touching the cap with a finger; (5) touch the strip to the cap, then remove the strip: the leaf stays at about 30 degrees. Learners record leaf divergence in degrees in a table: neutral 0 degrees, strip near (not touching) about 30 degrees, after touching and removing about 30 degrees, after finger touch 0 degrees. Then, in a quick demonstration, Mr Mwangi holds a rubbed balloon (dry conditions) 2 cm above torn paper pieces: they jump to the balloon, and some fall	Model the first trial slowly while the groups watch. Say: "Eyes on the leaf, not the strip. Read the angle at the protractor scale and record it in degrees." Circulate and ask each group: "What is the leaf doing now that the strip has gone?" Stress the difference between near but not touching, and touching. If a group's leaf falls quickly, say: "Check that your strip and the electroscope are dry, and that you rubbed for 20 strokes." Remind learners that the leaf is fragile and must not be breathed on. For the paper demonstration say: "Watch a paper piece. It jumps up, touches, and then... what happens?" WAIT TIME: after the demonstration, wait 10-15 seconds in silence before taking observations. Cold-call 4 groups to report one measurement each, and record the class data on the board in a table.	Structured data collection with a comparison table: recording leaf divergence in degrees under different conditions (near, touching, finger earthing) lets learners see patterns. They notice that touching leaves the leaf at about 30 degrees, the same as when the strip was near, but the charge stays after the strip is removed. Linking the electroscope result to the paper pieces moves learners from a lab observation to the anchoring phenomenon.	Check that group tables show: neutral 0 degrees; strip near about 30 degrees; after touching and removing about 30 degrees; after finger 0 degrees. Observable indicators: learners read angles in degrees (not "high" or "low"); learners notice the difference between the leaf falling back when the strip is only near and staying up after contact; learners state in their own words that the paper pieces that fell off had touched the balloon. Ask 3 learners: "What is different between near and touching?" and listen for "the charge stayed" or "something moved onto the electroscope".

	off again after touching it. Learners record which pieces fall off, and when.			
Explain Phase  VIDEO: Pith ball electroscope <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Electric Conductors and Insulators Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners work in their groups to explain the observations. Using Lesson 4 (conductors have free electrons) and Lesson 3 (friction transfers electrons), they answer: (1) Where did the extra electrons on the polythene strip come from? (2) When the negative strip touched the metal cap, what could the free electrons do? (3) Why does the leaf stay at about 30 degrees after the strip is removed? (4) What sign of charge does the electroscope now have? Each group draws a particle-level diagram in three steps: before contact (neutral cap, stem and leaf), during contact (electrons moving from the strip onto the cap and spreading down to the leaf), and after (electroscope with excess electrons, leaf and stem repelling each other). Learners then use the rule: contact gives the SAME sign as the charged object. They apply it to the paper pieces: a neutral paper piece is first attracted to the negative balloon, touches it, gains some electrons, becomes negative, and is then repelled and falls off. Kipchoge asks whether the cap gets positive because the leaf moves: the class uses the repulsion test, with the leaf rising further when a negative object is brought near, to answer.	Say: "Use only electrons moving. Protons stay in the nucleus." Ask: "When the strip touched the cap, which way did the electrons move, and why did they spread to the leaf?" WAIT TIME: give 10-15 seconds of silent thinking before groups discuss. Then ask learners to turn to a partner for 2 minutes. Cold-call 3 pairs to explain one step each (before, during, after). Correct any phrase like "positive charge moved" with: "In a solid, only electrons move. Say it again using electrons." Ask: "Why do the stem and leaf both become negative? What do like charges do?" Use the answer "the leaf and stem repel each other, so the leaf rises". Then ask: "Now think about the paper piece that touched the balloon. What sign does it have now, and what happens?" Finish by writing the rule on the board: "Contact gives the SAME sign. Electrons are transferred, not created." Ask: "If we count the extra electrons on the balloon, about 6×10^9 (Lesson 3), does the balloon lose any when the paper touches it?" Guide them to say: a few move to the paper, and total charge is conserved.	Claim-Evidence-Reasoning with particle-level modelling: learners make the claim 'the electroscope gained the same (negative) sign', support it with the 30 degrees leaf divergence that remains after removal, and give reasoning with free electrons and like charges repelling. Drawing a three-step before/during/after diagram makes the invisible electron transfer visible and transfers to the paper pieces.	Look for diagrams that: show electrons (not protons) moving from the strip to the cap; show extra electrons spread to the leaf and stem; label the electroscope as negative afterwards; use the words 'transfer' and 'same sign'. Listen for correct reasoning that the leaf rises because the like (negative) charges repel. Common errors to note: 'the strip's charge was used up', 'positive charges moved', or 'the electroscope became positive'. Ask an exit question: "A positive perspex rod touches a neutral electroscope. What sign does it gain, and which way do electrons move?" Expect: positive, electrons move from the electroscope to the rod.
Driving Question Board (DQB) Creation  VIDEO: 02 Conductors Insulators And	Learners return to the class Driving Question Board started in Lesson 1. Working in groups, they write on sticky notes (a) one question they can now answer, with evidence from today (for example, 'Why did the paper	Say: "Look at our board. Which of our questions can we now answer with evidence from today?" Give each group 3 minutes to write notes. WAIT TIME: wait 10-15 seconds after asking for a new question so that learners think	Driving Question Board with 'Answered / Partly answered / New questions': sorting questions against evidence shows learners how their understanding is growing and gives them ownership of the investigation. It keeps the unit	Check that each group's answered note cites evidence (such as 'leaf stayed at about 30 degrees after the strip was removed') and not just a statement. Check that new questions are testable or linked to the phenomenon. Indicators of

Semi Conductors <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	pieces fall off the balloon? Because they touched and gained the same negative charge, and like charges repel'), and (b) one new question that today's evidence raises (for example, 'Why is the neutral paper attracted to the balloon in the first place?', 'What if the electroscope were touched with a finger?' or 'How can we get the OPPOSITE sign on a conductor?'). They place the notes on the DQB under the headings 'Answered', 'Partly answered' and 'New questions'. Amina adds a note linking humidity to the charge leaking away from the electroscope.	before writing. Cold-call 3 groups to read a note aloud and say where it should go. Point out the questions about attraction of neutral paper and the wall, and say: "We will return to these in the next lesson, when we look at charging by induction and earthing." Move the question 'Where do electric charges come from?' to 'Partly answered' and ask: "What do we know now about where the charge comes from?"	coherent from lesson to lesson and motivates Lesson 6.	progress: at least 3 notes use the words electrons, transfer or same sign correctly; at least 1 group asks about the attraction of neutral objects, which leads to induction in Lesson 6.
Model Building Phase  VIDEO: 07 Conductors And Insulators <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria Search ARES for similar videos  HTML: Electric Conductors and Insulators (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners take out the particle-level model of the balloon, hair, wall and paper they have been building since Lesson 1. Individually, they revise it by adding: (1) a paper piece near the negative balloon being attracted; (2) the same paper piece touching the balloon, with a few electrons shown moving from the balloon onto the paper; (3) the paper piece now negative and repelled, falling off. They label the balloon as negative (excess of electrons), the hair as positive (deficit of electrons), and use arrows for electron movement and for the electrostatic force. They add a short caption: 'Contact gives the SAME sign. Electrons are transferred, not created.' Then they complete a short summary of the model changes with a partner.	Say: "Revise your model from last lesson. Add today's idea: what happens when the paper touches the balloon?" Circulate and prompt: "Which particles moved? Which way? What sign does the paper have afterwards? What does the paper do next?" Insist that balloon is negative and hair is positive. WAIT TIME: allow 10-15 seconds of silent thinking before learners start to draw. Cold-call 3 learners to explain one change in their model, and ask a different learner to say whether they agree and why. Ask: "What is still missing from our model?" Guide them to say that it does not yet explain why the NEUTRAL paper was attracted before touching, which is Lesson 6. Collect the models to check.	Iterative model revision with peer critique: learners add today's evidence to the same model rather than starting again. Revising the model shows how scientific explanations change with new evidence, and peer explanations check that the model really accounts for the phenomenon.	Check that revised models: show electrons (not protons) moving between balloon and paper; label the paper as negative after contact; show repulsion as the reason for falling off; keep the balloon negative and hair positive; and show gaps honestly (attraction of neutral paper unexplained). A learner meets the target if they can say, using the model: 'The paper touched the balloon, gained electrons, became negative like the balloon, and was repelled.' Record who needs support with direction of electron movement.

D. TEACHER REFLECTION

Which learners still believed that charge disappears when the strip is removed or that positive charge moves in solids, and how did the three-step particle diagram

change their thinking? Did the electroscope result (about 30 degrees remaining after contact) create a real surprise for the class, and did learners connect it to the paper pieces that fell off the balloon? Were dry conditions kept, and if some groups' leaves fell quickly, did that lead to a useful humidity discussion (Mombasa, about 80%, compared with Nairobi, about 35%)? Which questions on the DQB should be used to open Lesson 6, and who needs more support to contrast the SAME sign (contact) with the OPPOSITE sign (induction) next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	The leaf of a neutral electroscope stayed at about 30 degrees after a negative polythene strip touched the cap and was removed (0 degrees before; about 30 degrees also when the strip was only near). Touching the cap with a finger returned the leaf to 0 degrees. Paper pieces jumped up to the rubbed balloon and some fell off again after touching it.
What did I learn?	A conductor touched by a charged object gains the SAME sign of charge by transfer of electrons. Free electrons in the conductor let the charge spread to the leaf, and like charges repel, so the leaf stays raised. Charge is transferred, not created.
How does this explain the phenomenon?	The negative strip passed extra electrons to the cap, stem and leaf, so they all became negative and repelled each other. In the same way, a paper piece that touches the negative balloon gains electrons, becomes negative, and is repelled, so it falls off. This adds one more step to the balloon-and-hair explanation, though why neutral paper is attracted at first is still to be explained.

LESSON 6 (80 minutes (2 x 40-minute periods)): Charging by Induction and Earthing: Charging Without Touching

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explain how a charged object can attract a neutral object by induction, and show how a conductor can be given the OPPOSITE sign of charge using induction and earthing. This explains why the neutral wall and paper are attracted to the negative balloon.
Knowledge	Induction is the separation of charge in a neutral object when a charged object is brought near without touching. In a conductor, free electrons move away from a negative object (or toward a positive one), leaving the near side with the opposite sign. In an insulator such as paper or a painted wall, electrons shift only slightly within their atoms. Attraction is stronger than repulsion because the opposite charge is nearer. Charging by induction needs earthing and gives the OPPOSITE sign to the charged object (contact gives the SAME sign). Earthing lets electrons flow to or from the Earth.
Skills	Predicting and observing with a gold-leaf electroscope; measuring leaf divergence in degrees; developing and using particle-level models (NGSS SEP 2); analysing and interpreting data (SEP 4); constructing explanations (SEP 6); using evidence to compare contact and induction.
Attitudes	Curiosity about invisible forces; honesty in reporting observations, including surprises; careful and safe handling of apparatus; respect for peers' ideas during discussion; persistence when explaining a puzzling observation.
Key Inquiry Question	How can we charge conductors? Where do electric charges come from? Why does a rubbed balloon attract or repel other objects?
Purpose in Storyline	Lessons 3 to 5 showed that friction and contact transfer electrons, and that contact gives the SAME sign. This lesson answers an open question from Lesson 1: why is a NEUTRAL wall or paper attracted to the balloon, and why is it never repelled? Learners meet induction and earthing, and add them to the model. This prepares for Lesson 7 (distribution of charge) and Lesson 8 (using the electroscope to detect the sign).
Safety Notes	Use only dry, low-voltage materials. No mains electricity. Keep strips and rods away from faces. Handle the gold-leaf electroscope gently because the leaf is fragile and tears easily. Do not let learners touch the cap except when instructed. Keep the room dry: store balloons and strips in a dry place and wipe them before use. A small spark may be felt on touching, which is harmless but should be mentioned beforehand.

B. LESSON OVERVIEW

Mr Mwangi opens by asking why the neutral wall and paper pieces are attracted to the balloon but never repelled. Learners first predict. They then observe three pieces of evidence: (1) paper pieces and a wall near a negative balloon, (2) a neutral electroscope with a rubbed polythene strip brought near the cap but not touching, where the leaf rises to about 30 degrees and falls back to 0 degrees when the strip is removed, and (3) a four-step induction with earthing in which a finger touch returns the leaf to 0 degrees and the conductor ends with the OPPOSITE sign. In the Explain phase, learners use particle diagrams to show why attraction is stronger than repulsion and contrast contact with induction. They update the Driving Question Board (DQB) and revise their model of the balloon, hair, wall and paper by adding charge separation in the wall and paper.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Separation and Transfer of Charge - Overview Source: CK-12  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: electrostatics induction charge separation	Mr Mwangi holds a rubbed balloon (negative) 2 cm above torn paper pieces (about 1 cm x 1 cm). Pieces jump up to the balloon. Learners recall Lesson 1: the unrubbed balloon did nothing. Individually, learners write and sketch a prediction in their notebooks: 'The paper is neutral. It has no charge. So why does it jump to a charged balloon? What do you predict will happen to a neutral electroscope when a negative strip is brought near the cap but NOT touching it?' They draw the leaf at the predicted angle. They also predict: 'If I then touch the cap with my finger while the strip is still near, what will happen to the leaf?' They share with a partner, then write their ideas on sticky notes for the DQB.	Say: 'Wanjiru, Otieno, Amina and Kipchoge all saw the paper jump. Nobody saw it repelled. Let's think about why.' Ask: 'Is the paper charged before the balloon arrives?' Give WAIT TIME of 10-15 seconds in silence. Cold-call 3 learners (including Amina and Otieno) for their ideas without evaluating them. Say: 'I will not tell you if you are right yet. Draw what you think the electrons in the paper are doing.' Walk around and note common ideas, such as 'the paper must have gained charge' or 'the balloon is sticky'. Remind the class of the Lesson 5 contact result: a conductor touched by a charged object gains the SAME sign. Ask: 'Is the paper touching the balloon when it first jumps?' Record 3 predictions about the electroscope on the board with no judgement.	Predict-Observe-Explain (POE) with individual prediction first: committing to a written prediction creates a cognitive tension that makes learners notice the evidence. Surfacing initial ideas, such as 'paper must be charged', lets the teacher target them in later phases.	Check notebooks for: (a) learners who say the paper is already charged, (b) learners who predict the leaf stays at 0 degrees, (c) learners who predict the leaf rises, (d) any learner who mentions electrons moving without contact. Tally on a clipboard how many predicted the leaf would rise (target: identify the groups who need extra support in the Observe phase).
Observe Phase  VIDEO: Separation and Transfer of Charge - Overview Source: CK-12  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: electrostatics induction charge separation	In groups of 4, learners rotate through three stations (about 12 minutes each). Station 1, Neutral objects near a charged balloon: A balloon is rubbed on dry hair for 20 strokes (about 10 seconds). Learners hold it about 2 cm above paper pieces, then place it against a painted wall for about 3 minutes (range 2 to 5 minutes) and record what happens. Learners also test an unrubbed balloon (no effect). They record whether the paper is ever repelled before it touches, and what happens to pieces that touch the balloon and fall off. Station 2, Induction on the electroscope: A neutral gold-leaf electroscope has the leaf at 0 degrees. A rubbed polythene strip (negative, rubbed with wool/fur) is	Before the stations, say: 'Your job is to collect evidence. Write down exactly what you see, with angles in degrees. Do not explain yet.' Model reading the leaf angle against the scale. Circulate and ask each group: 'What happened to the leaf when the strip was brought near but not touching? Where did the strip never touch?' Give WAIT TIME of 10-15 seconds before accepting answers. At Station 3, say: 'Step order matters. The finger comes off BEFORE the strip. Why do you think that is?' Do not answer for them. Ask Kipchoge's group: 'You felt a tiny tingle when your finger touched the cap. What do you think moved?' Remind learners that only electrons move in solids. Cold-call	Data table with anomalous-result discussion: recording the same measurements (angles in degrees) at each step lets learners compare induction (no contact, leaf falls back) with induction plus earthing (leaf stays up) and with contact (Lesson 5, same sign). Seeing the numbers side by side makes the pattern visible before any explanation is given. This builds the SEP of analysing and interpreting data.	Observable indicators: learners (a) read the leaf angle within about 5 degrees of 30 degrees and 0 degrees, (b) record all four steps in the correct order, (c) note that the leaf falls to 0 degrees when the strip is removed in Station 2 (no charge left), but stays at about 30 degrees after Step 4 in Station 3 (charge left), (d) say that the strip never touched the cap. Use a checklist with the 4 steps; groups who miss Step 3 are asked to redo it.

	brought near the cap WITHOUT touching: the leaf rises to about 30 degrees. The strip is removed: the leaf falls back to 0 degrees. Learners record the angles in a table and answer: 'Did any charge stay on the electroscope?' Station 3, Charging by induction with earthing (Mr Mwangi demonstrates this one while groups watch, then pairs try it in 4 steps): Step 1: Bring the negative strip near the cap, without touching: the leaf rises to about 30 degrees. Step 2: While the strip is still near, touch the cap with a finger (earthing): the leaf falls to 0 degrees. Step 3: Remove the finger FIRST, with the strip still near: the leaf stays at 0 degrees. Step 4: Remove the strip: the leaf rises to about 30 degrees and stays up. Learners record the angle at each step. Sign test: bring the negative strip near again, and the leaf rises further (above 30 degrees), which shows the electroscope is now positively charged. (Learners are told that the leaf rises above 30 degrees when a negative object is brought near an electroscope that is already negative; for a positive electroscope the effect is the opposite, so they compare this with the earlier data and discuss in the Explain phase.) Learners also compare this with Lesson 5, in which touching with the strip left the leaf at about 30 degrees with the SAME sign.	4 learners across the room to describe one reading each (for example, 'leaf at about 30 degrees with strip near') and record these on the class table. If humidity is a problem, wipe the strip with a dry cloth and remind the class: 'Teachers use dry conditions. Humidity lets charge leak away, as in Mombasa in the rainy season.' Check that learners are only touching the cap when instructed.		
Explain Phase  VIDEO: Pith ball electroscope <i>Related topic: not an exact match for this lesson.</i>	Learners work in groups of 4 to build explanations from their evidence. Task 1, Charge separation: Using a large sheet, each group draws a particle-level diagram of the neutral electroscope (cap, stem, leaf) as	Say: 'Use only electrons as the movers. Protons stay in the nucleus. Never write that positive charge moved.' Circulate and ask: 'Where did the electrons in the electroscope go when I touched the cap?' WAIT TIME of 10-15	Model-based reasoning with Claim-Evidence-Reasoning (CER): learners build a particle-level model of electron movement and use their angle data as evidence to justify claims about the sign of charge. Comparing contact and	Indicators of understanding: (a) diagrams show electrons, not protons, moving; (b) the near side of the paper is labelled positive and the far side negative; (c) learners state that the electroscope ends positive after

Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: electrostatics induction charge separation	the negative strip is brought near the cap. They show the free electrons in the metal (Lesson 4: conductors have free electrons) being repelled down to the leaf and stem, leaving the cap positive. Both leaf and stem become negative and repel each other, so the leaf rises to about 30 degrees. The total charge is still zero. When the strip is removed, the electrons spread back, and the leaf falls to 0 degrees. Task 2, Earthing and the opposite sign: Groups add the finger step. With the strip near, electrons are repelled through the finger to the Earth. The electroscope now has a deficit of electrons. When the finger is removed and then the strip is removed, the remaining deficit spreads over the cap and leaf, so the leaf rises to about 30 degrees. The electroscope is positive, the OPPOSITE of the strip. They explain why the order (finger off first) matters: otherwise electrons flow back from the Earth. Task 3, Contrast with contact: Groups complete a two-column comparison of contact (Lesson 5: same sign, strip touches) and induction (opposite sign, strip never touches, earthing needed). Task 4, Back to the phenomenon: Groups apply the idea to the wall and paper. The negative balloon repels electrons in the neutral paper, so electrons shift slightly away, leaving the near side positive and the far side negative. The nearer positive side is attracted more strongly than the farther negative side is repelled, so there is a net attraction. In an insulator such as paper or a painted wall, the electrons shift only slightly within their atoms, but	seconds. Cold-call 3 groups to explain Task 2 using their diagrams. Probe: 'If the electrons left, why is the electroscope positive and not neutral?' Prompt for the term 'deficit of electrons'. Ask Amina's group: 'Does induction explain why the balloon is attracted to the wall in Mombasa on a rainy day?' Guide them to the humidity idea: water vapour lets charge leak away from the balloon, so there is less to induce. Ask: 'Why is a neutral object attracted but never repelled?' and wait for answers about the nearer charge. Correct misconceptions directly: 'The paper does not become permanently charged by induction. It only separates charge.' Use the phrase 'Induction gives the OPPOSITE sign. Contact gives the SAME sign.' and have the class repeat it. Select 2 groups to present diagrams, then ask the class to critique them using the evidence ('Which data supports this?').	induction in a two-column table helps them to separate two ideas that are easily confused. Applying the model to the paper and wall transfers the idea back to the anchoring phenomenon.	induction with earthing; (d) learners say that attraction is stronger because the opposite charge is closer; (e) learners correctly say contact gives the same sign and induction the opposite. Use a 3-question exit check: 'What sign does the electroscope have after Step 4? Why was the finger removed first? Why is paper attracted but not repelled?' Learners who say the paper is 'charged' or that protons move are given a short follow-up conversation.
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	the effect is the same. This is why the balloon sticks to the wall and the paper jumps, and why a neutral object is attracted but never repelled. Pieces that touch the balloon may gain a little electrons and the same sign, and then fall off. They finish with: 'Where do the charges in the paper come from?' (They were there all along; they just moved a little. No charge was created.)			
Driving Question Board (DQB) Creation  VIDEO: Separation and Transfer of Charge - Overview Source: CK-12  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: electrostatics induction charge separation	Learners return to the class DQB from Lesson 1. In groups, they review the questions on it and the sticky notes from today's Predict phase. They (1) move or mark questions now answered, for example 'Why is paper attracted to the balloon if it is neutral?' and 'Why does the balloon stick to the wall?'; (2) write the answer on a new card with the evidence (angles of 30 degrees and 0 degrees; the 4 steps); and (3) add new questions that arise, such as 'Why does the electroscope leaf rise above 30 degrees when a negative strip is brought near?', 'How do we know the sign of a charge?', 'Does charge spread out evenly on a conductor?', 'Why does the shock from a woollen jersey happen?' and 'Is the shock like lightning, as Kipchoge asked?'. Each group places at least one new question on the board.	Say: 'Look at the board. Which questions can we now answer with evidence from today?' Give WAIT TIME of 10-15 seconds. Cold-call 3 learners to read a question and say whether it is answered, partly answered or still open. Ask Otieno to link his observation of the balloon on the classroom wall to today's explanation. Ask Kipchoge to restate his question about lightning and say: 'We will return to this in Lesson 7 when we look at where charge goes on a conductor and sharp points.' Group the new questions under the KICD inquiry questions (for example 'How can we detect and measure electric charge?'). Ensure each answered question has evidence on its card, not just an opinion.	Driving Question Board as a public record of growing understanding: separating answered, partly answered and open questions shows learners that science is cumulative and that each lesson links to the next. Writing evidence on each answer card reinforces evidence-based reasoning.	Indicators: learners' answer cards cite specific data (for example 'leaf stays at about 30 degrees after removing the strip'); new questions are testable and connected to charge sign, distribution or discharge; at least 80% of groups add one new question. Note which questions are most common to guide Lessons 7 and 8.
Model Building Phase  VIDEO: Separation and Transfer of Charge - Overview Source: CK-12	Learners take out their model from Lesson 5: a labelled particle-level diagram of the balloon, hair, wall and paper. Individually, they revise it by (1) showing the neutral wall and paper near the negative balloon with electrons shifted away, so the near surface is positive and the far side negative;	Say: 'Your model must now explain the wall and the paper. Show me electrons, not protons, moving.' Circulate and ask: 'Where are the electrons in the paper now? Why is the near side positive?' WAIT TIME of 10-15 seconds. Cold-call 3 learners to explain one change they made.	Cumulative model revision (SEP 2: developing and using models): revising the same particle-level diagram each lesson shows how new evidence refines explanations. Adding induction to the model links charge separation to the anchoring phenomenon. Noting what the model cannot yet	Indicators on the revised model: (a) electrons shifted away from the balloon in the wall and paper; (b) near side positive and far side negative, with the near attraction larger; (c) no protons moving; (d) a statement that charge is conserved; (e) humidity noted as a cause of leakage; (f) one

Search ARES for similar videos READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: electrostatics induction charge separation	(2) adding arrows for the electrostatic force: a stronger attraction between the balloon and the nearer positive charges, and a weaker repulsion from the farther negative charges; (3) labelling that no charge was created and total charge is conserved; (4) adding a note on the effect of humidity: water vapour lets charge leak away, so the balloon falls quickly in Mombasa. They also add a small inset showing the electroscope with induction and earthing (4 steps), and write 2 to 3 sentences answering: 'Why is neutral paper attracted to the balloon but never repelled?' They colour-code electrons and use a different colour for the changes made today. Finally, they note one thing their model still cannot explain, such as the spark from a woollen jersey in Nairobi.	Prompt struggling learners: 'Is the paper charged overall, or just separated?' Ask the class: 'What is the difference between charge separation and gaining charge?' Check that all models show a balloon with an excess of electrons (negative) and hair with a deficit (positive), as in earlier lessons. Collect models for review. Preview Lesson 7: 'Next, we ask where the charge goes on a conductor. Could that help explain the shock Wanjiru and Kipchoge felt?'	explain builds metacognition and motivates the next lesson.	unanswered question listed. Use a 4-point checklist; learners who miss (a) or (c) are revisited at the start of Lesson 7.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners clearly separate induction (opposite sign, earthing needed, no touching) from contact (same sign) in Lesson 5? Which diagrams still show protons moving? (2) Did the step order with the finger (remove finger before the strip) cause confusion? If so, plan a short re-demonstration. (3) Did dry conditions hold? Note any groups whose leaf did not rise to about 30 degrees and check the strips, cloth and room humidity. (4) Were learners able to apply the idea to the neutral paper and wall, and not just the electroscope? (5) Which learners (for example Amina with the humidity idea and Kipchoge with the lightning idea) need to be heard more? (6) Which new DQB questions should shape Lesson 7?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A neutral electroscope with a negative polythene strip brought near the cap, not touching: the leaf rose to about 30 degrees and fell back to 0 degrees when the strip was removed. With the strip near and a finger touching the cap (earthing), the leaf fell to 0 degrees. After removing the finger and then the strip, the leaf rose to about 30 degrees and stayed up. Neutral paper pieces jumped to the negative balloon and the balloon stuck to the wall, with no repulsion seen.
What did I learn?	A charged object can separate charge in a neutral object without touching it (induction). Earthing lets electrons flow to or from the Earth, and charging by induction with earthing gives the OPPOSITE sign, while contact gives the SAME sign. Only electrons move, and charge is not created but transferred.
How does this explain the	The negative balloon repels electrons in the neutral paper and wall, leaving the near surface positive and the far side negative. The

phenomenon?	nearer positive charge is attracted more strongly than the farther negative charge is repelled, so a neutral object is attracted but never repelled. This explains why the paper jumps and the balloon sticks to the wall. On humid days, such as in Mombasa, the balloon's charge leaks away, so the effect is weak or short-lived.
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LESSON 7 (80 minutes (2 x 40-minute periods)): Distribution of Charge on Conductors: Outer Surfaces, Points, Sparks and Lightning Conductors

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate where charge sits on a charged conductor, and use point discharge to explain sparks such as the Nairobi jersey shock and the action of lightning conductors. This moves the class closer to answering: Where do electric charges come from, and why does a rubbed balloon attract or repel other objects?
Knowledge	By the end of the lesson, learners should be able to state that (1) charge on a conductor collects on the outer surface (hollow can: outside about 40 degrees, inside 0 degrees), (2) charge is concentrated at points (pointed end about 45 degrees, flat end about 20 degrees), and (3) point discharge, in which a strong field at a point ionises the air and charge leaks away, explains sparks and lightning conductors.
Skills	Planning and carrying out investigations (using a proof plane and electroscope); analysing and interpreting data (comparing leaf divergence in degrees); constructing explanations (claim-evidence-reasoning); developing and using models (revising the particle-level diagram); communication and collaboration.
Attitudes	Curiosity about invisible forces, respect for evidence, care and safety when handling charged apparatus, and appreciation of how science protects buildings and people in lightning-prone areas such as Kericho and Kisumu.
Key Inquiry Question	Where does charge go on a conductor? (KICD Key Inquiry Question 5), linked to the driving question: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects?
Purpose in Storyline	Lesson 4 showed that conductors have free electrons that move through the material, and Lesson 5 showed that a conductor touched by a charged object gains the same sign of charge by electron transfer. Lesson 6 introduced earthing. Lesson 7 asks where the transferred electrons end up on a conductor, and how this explains the spark Kipchoge asked about (Is the shock like lightning?). It prepares Lesson 8 (detecting and measuring charge with the electroscope) and Lesson 9 (applications, including lightning conductors).
Safety Notes	Mr Mwangi uses only dry apparatus in a dry room and does all charging himself (small charges of about nC only). Learners must never touch a real lightning conductor or go near one in a storm, and should shelter indoors during thunderstorms, away from open fields, trees and Lake Victoria shores. Handle the fragile gold leaf gently, and keep pointed metal away from faces and eyes. Do not run sparking demonstrations near flammable materials or fuel.

B. LESSON OVERVIEW

Mr Mwangi begins with Kipchoge's question from the Nairobi dry-season day: is the jersey shock like lightning? Learners predict where the extra electrons will sit on a charged metal can and on a pear-shaped conductor. They then use a proof plane and a gold-leaf electroscope to collect data: outside of the can gives about 40 degrees and inside gives 0 degrees, while the pointed end gives about 45 degrees and the flat end about 20 degrees. They explain the pattern with repulsion between mobile electrons (from Lessons 4 and 5), use point discharge to explain sparks and lightning conductors in Kericho and Kisumu, and revise the DQB and their particle-level models of the balloon, hair, wall and paper to include charge spreading out and leaking from points.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 07 Lightning Arrestors, Lightning Conductor Source: Seavuria  Search ARES for similar videos  HTML: Charges and Fields <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Wanjiru recalls how her hair stands up after combing on dry Nairobi mornings, and Kipchoge asks, Is the shock from a woollen jersey like lightning? Learners look at a hollow metal can on an insulating stand and a pear-shaped metal conductor. Individually, then in pairs, they draw each object and use dots (for extra electrons) to predict where the extra electrons will sit if each conductor is given a negative charge from a rubbed polythene strip. They also predict which spot will make an electroscope leaf rise higher when touched with a proof plane: outside or inside the can, and the pointed end or the flat end. Each pair writes a one-sentence reason using what they know about free electrons in conductors and like charges repelling.	Say: 'Kipchoge asked whether our jersey shock is like lightning. Today we find out where charge goes when it sits on a metal object.' Show the can and the pear-shaped conductor. Say: 'Draw the object. Put dots where you think the extra electrons will be. Write why.' Give 2 minutes of silent individual drawing. WAIT TIME: 10-15 seconds after asking, 'Where would 10 electrons push each other to, if they can move freely?' Cold-call 4 learners to share predictions without saying whether they are right. Record the predictions in tally form on the board (outside, inside, spread evenly, at points). Ask Amina, Otieno and Wanjiru to explain their reasoning to the class. Say: 'We will test every prediction with data. Keep your drawing; we will revise it.' Do not give the answer yet.	Predict-Observe-Explain with a individual-to-pair drawing task: committing to a prediction and reasoning before evidence makes later conflict between the prediction and the data visible, which drives conceptual change. Learners reuse the Lesson 4 and 5 idea of free electrons in conductors.	Check the drawings: (1) Do learners draw dots as electrons (negative), not as positive charges moving? (2) Does the reasoning mention that electrons are free to move and repel each other? (3) Tally the number of learners predicting outside, inside, even and points. Look for the misconception that charge spreads evenly through the inside of the metal, and note which learners use words such as 'repel' or 'spread out'.
Observe Phase  VIDEO: 06 Lightning Arrestors, Electrostatic Shielding Source: Seavuria  Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners watch Mr Mwangi's demonstration in dry conditions and, in groups of 4, record and compare data. Part A: The hollow metal can is charged negatively using a rubbed polythene strip (contact charging, Lesson 5). A proof plane is touched on the OUTSIDE of the can, then touched to the cap of the gold-leaf electroscope; the leaf rises to about 40 degrees. The electroscope is discharged by earthing with a finger (Lesson 6), returning the leaf to 0 degrees. The proof plane is then touched on the INSIDE of the same can and transferred to the electroscope; the leaf stays at 0 degrees. Part B: The pear-shaped (pointed) conductor is charged. The proof	Say: 'Watch the leaf, not my hands. Record the angle in degrees.' Before each touch, ask: 'What do you predict for this reading?' Make learners say a number or a range. WAIT TIME: 10-15 seconds after each reading before anyone speaks. Ask: 'What changed between outside and inside?' Cold-call 3 groups for each part. Ensure the electroscope is earthed to 0 degrees before each new reading: 'Why must I touch the cap with my finger before the next test?' (Lesson 6 earthing). Draw the results table on the board and fill it in as: outside 40, inside 0, pointed end 45, flat end 20. Say: 'A proof plane picks up a small sample of charge from the spot it touches. A higher	Data table and pattern-finding with a gallery of numbers: comparing leaf angles across locations (40 vs 0, 45 vs 20) lets learners see a clear pattern through quantitative evidence. Using a proof plane as a probe shows an invisible quantity by a visible measure (leaf divergence in degrees).	Observable indicators: (1) Learners correctly record 40 and 0 degrees (outside and inside), and 45 and 20 degrees (pointed and flat ends) with units. (2) Learners state that a bigger leaf angle means more charge was collected. (3) Learners remember to earth the electroscope between tests. (4) In their explanations, learners connect inside = 0 degrees to 'no charge on the inside surface', not 'no electrons inside the metal'. Note any group that claims the can is not charged because the inside reads 0.

	plane touched at the pointed end gives about 45 degrees; after earthing the electroscope to 0 degrees, the flat end gives about 20 degrees. Part C: The class discusses the Nairobi spark: what happens to the crackle in a dark room after removing a woollen jersey? Groups record a data table: location, leaf angle (degrees), and a sketch of where the charge is. Groups explain why the inside reading is 0 degrees even though the can is charged.	leaf angle means more charge at that spot.' Circulate and ask: 'Is it that the inside has no electrons, or that the extra electrons are not there?' Ask learners to link 'a rubbed balloon gains about 1 nC' to the idea that charge also sits on surfaces. Remind groups that the metal can is a conductor, unlike the balloon (an insulator), so its charge can move.		
Explain Phase  VIDEO: 07 Lightning Arrestors, Lightning Conductor Source: Seavuria  Search ARES for similar videos  HTML: Charges and Fields <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	In groups, learners write a claim-evidence-reasoning (CER) explanation answering: 'Where does charge go on a conductor, and why?' Claim: charge collects on the outer surface and is most concentrated at points. Evidence: outside 40 degrees vs inside 0 degrees; pointed end 45 degrees vs flat end 20 degrees. Reasoning: the free electrons in a conductor repel each other (like charges repel), so they move as far apart as possible, to the outer surface; at a sharply curved point they crowd more closely together. Then learners apply point discharge: at a point, the strong electric field ionises the air nearby, and charge leaks away into the air. They use this to explain (a) the Nairobi spark and faint crackle from a woollen jersey or comb on a dry day, and (b) lightning conductors on buildings in Kericho (tea country, frequent lightning) and Kisumu: a pointed metal rod on top of a building joined by thick copper strip to a buried earth plate. The pointed rod lets charge leak away gradually, and if lightning strikes it gives a safe path to the ground (earthing) instead of through the	Say: 'Use your numbers. A claim without evidence is just an opinion.' Provide CER sentence starters on the board. Ask: 'Why do free electrons end up on the outside?' WAIT TIME: 10-15 seconds. Cold-call 4 learners to complete the sentence 'The electrons repel each other so...'. Ask: 'Why would a point have more charge crowded together than a flat end?' Accept answers using the data and gently correct statements such as 'positive charges moved' (in solids only electrons move). Introduce the term 'point discharge': 'At a sharp point the field is so strong that it can ionise the air, and charge leaks away.' Show a picture or model of a lightning conductor in Kericho: pointed rod, thick copper strip, buried earth plate. Ask: 'Why a pointed rod and not a flat plate? Why thick copper? Why buried in the ground?' (point discharge, copper is a conductor, earthing from Lesson 6). Ask Kipchoge: 'Is the jersey shock like lightning?' Give the idea that both are discharges, differing in size. Do not use 'static electricity' as an explanation; use 'electrostatic	Claim-Evidence-Reasoning (CER) with transfer: connecting numerical evidence to a particle-level reason (electron repulsion) and then applying it to an unfamiliar real context (lightning conductors in Kericho, Kisumu) shows whether learners can use the idea rather than only recall it.	Look for: (1) claims that use 'outer surface' and 'points'; (2) evidence quoting both readings with degrees; (3) reasoning that includes repulsion between electrons and free movement in conductors; (4) correct use of 'discharge' and 'earthing'; (5) transfer: learners explain that the pointed rod leaks charge and the copper strip and earth plate carry the charge safely to the ground. Misconceptions to watch for: 'lightning conductors attract lightning', 'charge is created at the point', or 'positive charge flows through the copper'. Correct with 'electrons move, charge is transferred, not created'.

	building. Learners compare a lightning conductor with Kipchoge's question: is the jersey shock like lightning? They explain that both are discharges, with lightning a much larger one.	force' and 'discharge'. Hold a quick exit-check: 'Write one sentence linking the pointed end reading of 45 degrees to a lightning conductor.'		
Driving Question Board (DQB) Creation  VIDEO: 06 Lightning Arrestors, Electrostatic Shielding Source: Seavuria  Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners return to the DQB. They review which questions are answered by today's evidence: KICD question 5 (Where does charge go on a conductor?) and part of question 7 (How is electrostatics used in everyday life?). They add new questions on sticky notes, for example: 'Why does charge stay on the balloon but spread out on metal?', 'Can a lightning conductor stop lightning?', 'Why do fuel tankers trail a chain?', 'How does the electroscope tell us the sign of charge?' Learners place each note on the board under a category (answered, partly answered, new question) and write the evidence that answers it (e.g., outside 40 degrees, inside 0 degrees; pointed end 45 degrees, flat end 20 degrees). Amina asks whether humid coastal weather in Mombasa changes point discharge; this is added as a new question.	Say: 'Which of our questions can we now answer with evidence? Move the sticky note and write the evidence on it.' Ask 3 learners (cold-call) to move a note and justify it. WAIT TIME: 10-15 seconds before asking for new questions: 'What does today's evidence make you wonder next?' Collect at least 5 new questions. Point out that Lesson 8 will answer 'How can we detect and measure electric charge?' and Lesson 9 will answer 'How is electrostatics used in everyday life?' Praise questions that start with 'why' or 'how' and are testable. Link Amina's question back to humidity: 'Remember that on humid days water vapour lets charge escape; how might this connect to point discharge?'	Driving Question Board as a living record: moving questions to 'answered' with evidence helps learners see their own progress, and new questions keep the inquiry student-driven and coherent across lessons.	Check that notes are moved only with evidence (e.g., quoting 40 and 0 degrees, or 45 and 20 degrees). Count how many new questions are testable and linked to the phenomenon. Look for learners who connect Amina's humidity idea or Kipchoge's lightning question to today's data. Note learners who give no new question and follow up individually.
Model Building Phase  VIDEO: 04 Charge Distribution On A Spherical Conductor <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos	Learners revise their labelled particle-level diagram of the balloon, hair, wall and paper. In the previous lessons they showed electrons transferred from hair (positive) to balloon (negative), and induced separation of charge in the neutral wall and paper. Now they add: (1) the extra electrons on a charged conductor (such as a metal object touched by the balloon) are drawn spread on the outer surface, with closer spacing	Say: 'Add what is new in a different colour. Label everything. Use the word electrons for what moves.' Remind learners that the balloon is negative and hair is positive in every diagram. Circulate and ask: 'Where is the charge on the metal object in your model? Why there?' WAIT TIME: 10-15 seconds. Cold-call 3 learners to explain their revision. Ask: 'What does your model now explain that it could not explain before?' and 'What can it	Iterative model revision: adding new particle-level features each lesson, with colour-coded changes, shows how evidence changes explanation, and identifying the model's limits creates purpose for the next lesson.	Check models for: (1) electrons (not positive charges) shown on the outer surface of a conductor; (2) closer spacing at a point; (3) a point discharge inset with electrons leaking into air; (4) correct signs (balloon negative, hair positive); (5) evidence labels of 40, 0, 45 and 20 degrees; (6) a stated limitation of the model. Revisit learners whose models show charge inside the metal or evenly spread, and ask them to

 HTML: Charges and Fields <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	at any pointed part; (2) a note that on the insulating balloon the charge stays where it was rubbed, unlike on a metal conductor where it moves; (3) a small inset sketch of point discharge: electrons leaking from a point into the air, linked to the Nairobi spark and faint crackle from a woollen jersey, and to the lightning conductor in Kericho. They label the leaf angles (40 and 0 degrees; 45 and 20 degrees) beside the inset as evidence and write a two-sentence caption. They compare models with a partner and add one revision in a different colour.	still not explain?' Collect the answers (for example, how to measure the amount of charge, which is Lesson 8) and add them to the DQB. Tell learners to keep the models for Lesson 8.	point to the data that disagrees.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners explain the 0 degrees inside reading as 'no charge on the inside' rather than 'no electrons in the metal'? (2) How many learners used electron repulsion in their reasoning, and who still thought charge spreads evenly? (3) Did the dry conditions give clear readings of about 40, 0, 45 and 20 degrees, and did I remember to earth the electroscope between tests? (4) Did any learners say 'positive charge moved' or 'static electricity' as an explanation, and how did I respond? (5) Could learners connect the pointed end reading to the lightning conductors in Kericho and Kisumu and to Kipchoge's jersey shock question? (6) Which questions on the DQB still need evidence for Lesson 8 (detecting and measuring charge) and Lesson 9 (applications)? Plan follow-up for learners who need to rehearse the CER structure.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a charged hollow metal can gave an electroscope leaf reading of about 40 degrees when the proof plane touched the outside and 0 degrees when it touched the inside. On the pear-shaped conductor, the pointed end gave about 45 degrees and the flat end about 20 degrees.
What did I learn?	We learned that the free electrons on a charged conductor repel each other and move to the outer surface, and that charge is most concentrated at points. At a point, the strong field ionises the air and charge leaks away (point discharge), while earthing lets charge flow safely to the ground.
How does this explain the phenomenon?	This explains why a woollen jersey shock in dry Nairobi is a small spark (a discharge) and why lightning conductors, with a pointed rod joined by thick copper strip to a buried earth plate, protect buildings in Kericho and Kisumu. It also adds to our model of how electrons, not created charge, move and build up around the balloon, hair, wall and paper.

LESSON 8 (40 minutes (one double period of 80 minutes if the practical is done in groups)): Detecting and Measuring Charge: The Gold-Leaf Electroscope

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To let learners use a gold-leaf electroscope to detect charge, find its sign, discharge it, and relate leaf divergence to the amount of charge, so they can answer how we know that the balloon is charged and how much.
Knowledge	Learners explain that a gold-leaf electroscope detects charge because charge on the cap and leaf makes the leaf and stem repel (like charges repel). A larger leaf divergence means more charge (20 strokes gives about 30 degrees; 40 strokes gives about 50 degrees). A charged electroscope is used to find the sign of an unknown charge: an increase in divergence means the same sign, and a decrease means the opposite sign. Touching the cap with a finger (earthing) lets electrons flow to or from the Earth and returns the leaf to 0 degrees.
Skills	Developing and using models; planning and carrying out investigations; analysing and interpreting data (leaf divergence in degrees); constructing explanations using evidence; communicating scientific ideas clearly.
Attitudes	Curiosity about invisible forces; honesty in recording readings; careful teamwork and safe handling of apparatus; respect for evidence when it contradicts a prediction.
Key Inquiry Question	KICD Key Inquiry Question 6: How can we detect and measure electric charge? This also advances the Driving Question: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects?
Purpose in Storyline	Lessons 5, 6 and 7 showed how charge is moved onto conductors by contact, by induction and by earthing, and where it sits on a conductor. In Lesson 8 learners turn the electroscope from something they have used into a measuring tool. They gather evidence that the balloon really carries charge, that its sign can be tested, and that more rubbing gives more charge. This evidence feeds the final model in Lesson 9.
Safety Notes	Gold leaf is very thin and tears easily, so handle it only with the teacher's guidance and keep the electroscope case closed or away from draughts. Use only dry polythene strips, wool/fur and perspex rods. Keep the apparatus away from sinks and open windows because humidity makes charge leak away. Do not touch the cap during the readings. Nothing in this lesson uses mains electricity. Any spark felt is tiny and harmless, but learners should not rub and then touch each other's faces.

B. LESSON OVERVIEW

Mr Mwangi opens with Amina's claim that the balloon 'never works' at her coastal home in Mombasa, and Wanjiru's feeling of a shock after combing her hair. He asks, 'How could we tell, without relying on feelings, whether something is charged, and by how much?' Learners predict what a gold-leaf electroscope will do when a rubbed polythene strip is brought near the cap. In groups they then observe: a neutral electroscope with the leaf at 0 degrees; the leaf rising to about 30 degrees when a negative strip (20 strokes) is brought near without touching; the leaf staying at about 30 degrees after the strip touches the cap and is removed; the leaf rising to about 50 degrees with a stronger charge (40 strokes); the sign test, where a negative object raises the leaf above 30 degrees and a positive object lowers it below 30 degrees; and finger earthing returning the leaf to 0 degrees. In the Explain phase, learners use particle-level reasoning, with electrons only moving, to explain each result. They then add new questions to the Driving Question Board and revise their model of the balloon, hair, wall and paper by adding a 'how do we know?' detector box showing that the balloon's charge can be detected and compared.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Pith ball electroscopes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners sit in groups of four around one gold-leaf electroscope on a desk. They recall Wanjiru's hair standing up and Amina's balloon failing in Mombasa. Each learner draws a neutral electroscope (cap, stem, leaf) and predicts, with a sketch, what the leaf will do when a rubbed polythene strip (20 strokes with wool) is brought near the cap without touching it. They also predict what happens if the strip is rubbed with 40 strokes instead. Then they predict what will happen when the cap is touched with a finger. They write a prediction with a reason using the words 'electrons', 'repel' and 'leaf'.	Say: 'Last lesson we saw that charge crowds at points. Today's question: how can we detect charge and measure it, since we cannot see electrons?' Hold up the electroscope and point out the cap, metal stem and leaf. Show the leaf hanging down at 0 degrees. Say: 'Draw this neutral electroscope, then predict. Do not touch anything yet.' Give 2 minutes of silent individual writing. WAIT TIME: after asking 'What do you think the leaf will do when the negative strip comes near the cap?' wait 10-15 seconds before taking answers. Cold-call 3 learners, including one from a group that has not yet spoken, and ask each, 'What is your reason?' Record the three predictions on the board without judging them. Ask Amina to say how she would know whether her balloon in Mombasa was charged at all.	Predict-Observe-Explain (POE) with individual sketches: committing to a prediction and a reason before seeing data makes learners aware of what they already think about charge and electron movement, so the later observations either support or challenge those ideas. Learners' ideas are compared with the evidence in the Explain phase.	Check the sketches for these indicators: (1) the leaf drawn as lifting away from the stem; (2) the word 'repel' or 'same charge' used in the reason; (3) any mention of electrons moving onto or off the cap. Tally on the board how many learners predict 'leaf rises' versus 'no change'. Note who predicts that 40 strokes gives a bigger angle than 20 strokes. Note common ideas, such as 'the leaf is attracted to the strip', to revisit in the Explain phase.
Observe Phase  VIDEO: Pith ball electroscopes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Charges and Fields <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations	Learners work in groups of four with an electroscope, a polythene strip, wool/fur, a perspex rod, silk and a protractor or angle scale card placed behind the leaf. Step 1: Record the neutral leaf angle (0 degrees). Step 2: Rub the polythene strip with wool 20 strokes, bring it near the cap without touching it, and record the angle (about 30 degrees). Remove the strip and record what happens to the leaf. Step 3: Touch the strip to the cap (contact), remove the strip and record the angle (stays at about 30 degrees). Step 4: Sign test. With the electroscope now charged negatively at about 30	Distribute the apparatus and say: 'Handle the leaf gently. Take readings from the scale behind the leaf at eye level and do not touch the cap unless the instructions say so.' Move around the groups and ask: 'What happened to the leaf when the strip came near, and what happened when you took the strip away?' Ask each group to read out one measurement. Use cold-call: select 4 groups and ask each for the angle for one step, then compare their readings on the board (about 0, 30, 30, above 30, below 30, 0, 50). Prompt: 'Why do you think the angle is bigger with 40 strokes?' but do not give the	Data table with structured observation: recording angles in degrees for each action turns what learners see into comparable evidence. Comparing readings across groups shows which results are consistent (about 0, 30 and 50 degrees) and helps them see the pattern before an explanation is given.	Look for: (1) angles recorded within about 5 degrees of 0, 30 and 50 for the corresponding steps; (2) the sign test recorded correctly, with an angle above 30 for negative and below 30 for positive; (3) learners reading the scale at eye level; (4) earthing recorded as returning the leaf to 0 degrees. Use a quick tally of groups with correct results, and help groups whose strips were damp or rubbed too few times.

(English) Search ARES for similar readings	degrees, bring a second rubbed polythene strip (negative) near the cap and record the angle (rises above 30 degrees). Then bring a rubbed perspex rod (positive, rubbed with silk) near the cap and record the angle (falls below 30 degrees). Step 5: Earth the electroscope by touching the cap with a finger and record the angle (0 degrees). Step 6: Repeat the contact charging with 40 strokes and record the angle (about 50 degrees). Learners fill in a results table: Action, Number of strokes, Leaf divergence (degrees), Observation. They also watch a short teacher demonstration of the balloon (rubbed 20 strokes) brought near the cap, which makes the leaf rise, showing that the balloon from the phenomenon is charged.	answer yet. WAIT TIME: after the sign test say, 'Look at your table. What pattern do you see?' and wait 10-15 seconds in silence. Remind groups to keep the demonstration dry and to keep their hands away from the cap. Do the balloon demonstration at the front: rub the balloon 20 strokes on dry hair, hold it near the cap and show the leaf rising.		
Explain Phase  VIDEO: Conservation of charge <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	Groups use their results table to explain each result in terms of electrons. First they draw a particle-level diagram of the electroscope when the negative strip is near the cap: electrons in the metal are pushed down the stem to the leaf, so leaf and stem both have excess electrons and repel. They then explain contact: electrons move from the strip to the electroscope, so the leaf stays up at about 30 degrees even after the strip is removed (same sign as the strip). For the sign test they reason that adding more negative charge pushes more electrons into the leaf and stem, so repulsion and the angle increase; a positive rod attracts electrons up toward the cap, so the leaf and stem have fewer excess electrons and the angle falls. For earthing, they explain that electrons flow from the	Say: 'Use only electrons for movers. Protons stay in the nucleus.' Ask, 'Where did the electrons in the leaf come from when the strip touched the cap?' WAIT TIME: 10-15 seconds before calling on anyone. Cold-call 3 learners to explain one step each, then ask a fourth to challenge or add to the explanation. Correct 'the leaf is attracted to the strip' by asking, 'If the strip is removed and the leaf stays up, can the strip be the thing pulling it?' Ask, 'Why does a positive rod make the leaf fall, not rise? What is it doing to the electrons?' Point out that a leaf that rises only shows the object is charged and not the sign; a rise on an already charged electroscope shows the same sign, a fall shows the opposite sign. Note that a neutral object can attract by induction, so repulsion (a rise) is	Claim-Evidence-Reasoning (CER) with particle-level drawings: each group states a claim (for example, 'The electroscope now has excess electrons'), cites evidence from their table (for example, 'the leaf stayed at about 30 degrees after the strip was removed'), and gives reasoning using electron movement and like charges repelling. This links the measurements to the model and prevents memorising rules without understanding.	Listen for: (1) 'electrons move' rather than 'positive charge moves'; (2) the statement 'like charges repel' used to explain the leaf rising; (3) the sign test explained as an increase for the same sign and a decrease for the opposite sign; (4) earthing explained as electrons flowing to the Earth; (5) a link between bigger angle and more charge (50 degrees against 30 degrees). Use a 3-question exit check on mini whiteboards: 'What does a bigger angle show?', 'What does a decrease in angle show?', 'What does earthing do?' Count correct whiteboards and revisit the sign test if fewer than most are correct.

	electroscope to the Earth through the finger, leaving the electroscope neutral at 0 degrees. They compare 20 strokes (about 30 degrees) with 40 strokes (about 50 degrees) and write a statement: 'More charge on the leaf and stem gives more repulsion and a bigger leaf divergence.' They finish by relating this to the balloon: the rubbed balloon (negative, about 1 nC) caused the leaf to rise because it carries excess electrons.	the sure test. Ask Kipchoge to explain how the leaf's behaviour is like a small version of the spark he asked about. Summarise: 'Leaf divergence tells us the amount of charge, and the sign test tells us the sign.'		
Driving Question Board (DQB) Creation  VIDEO: Pith ball electroscope Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Charges and Fields <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners return to the class Driving Question Board and look at the questions from earlier lessons. Each learner writes one new question or answer on a sticky note. Possible new questions: 'How can we measure charge in coulombs and not only in degrees?', 'Why did the balloon stick for less time in Mombasa?', 'Can an electroscope detect charge from a lightning storm in Kisumu?', 'Why does the leaf slowly fall on a humid day?' Groups move sticky notes that have been answered (for example, 'How can we detect and measure electric charge?') to the 'Answered' section with a short evidence statement, such as 'The leaf rose to about 30 degrees with 20 strokes and about 50 degrees with 40 strokes.'	Say: 'Look at the board. Which questions can we now answer with our electroscope evidence?' Give 2 minutes for writing, then WAIT TIME: ask 'Which question should we move first and why?' and wait 10-15 seconds. Cold-call 3 learners to place a note and justify it using data. Ask Amina to restate her Mombasa question and ask the class how an electroscope could help test the humidity idea (the leaf would fall back to 0 degrees faster). Keep questions about applications, such as Kericho lightning conductors and fuel tankers in Nairobi, for Lesson 9.	Driving Question Board with evidence-based question sorting: asking learners to justify moving a question to 'Answered' using measurements keeps the focus on evidence and shows how the unit's questions are being resolved step by step. New questions keep learner curiosity driving the next lesson.	Check that each sticky note is a testable or clear question, that moved notes include a data-based justification (angle in degrees), and that at least a few new questions link detection to the balloon, humidity or lightning. Note which learners ask only recall questions and prompt them with 'What would you change to test it?'
Model Building Phase  VIDEO: Pith ball electroscope Source: Khan Academy (English - US curriculum)	Learners revise their labelled particle-level diagram of the balloon, hair, wall and paper from earlier lessons. They add a new 'How do we know?' box showing a gold-leaf electroscope near the negative balloon, with electrons drawn moving down to the leaf and stem and the leaf at about 30	Say: 'Add the new box to your model. Use arrows for electron movement and label every angle in degrees.' Circulate and ask, 'Where are the extra electrons in your diagram, and what is the leaf doing because of them?' WAIT TIME: ask 'What would your model show if the balloon were rubbed	Iterative model revision: adding the electroscope detector to the existing particle-level model connects measurable evidence (degrees) to the invisible electron transfer. Revising the same model across lessons shows how ideas build and become more complete.	Use a checklist: (1) electrons shown in the leaf and stem with correct sign; (2) leaf angle labelled in degrees with 30 and 50 linked to the number of strokes; (3) earthing shown as electron flow to the Earth; (4) balloon negative, hair positive, and wall and paper neutral still labelled correctly; (5) a

 Search ARES for similar videos  HTML: Coulomb's Law <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	degrees. They add arrows for the electrostatic force between leaf and stem, a note that 40 strokes (more electrons transferred) would raise the leaf to about 50 degrees, and an earthing path showing electrons flowing through a finger to the Earth. They keep the labels: balloon negative, hair positive, wall and paper neutral. Each learner writes two sentences under the model: one on how we detect charge and one on how we know its sign.	more?' and wait 10-15 seconds. Cold-call 3 learners to share one change to their model. Remind learners that charge is not created: electrons are transferred and total charge is conserved. Check that no diagram shows protons moving. Ask them to keep the model for Lesson 9, when they will write the final full explanation.		sentence that links larger divergence to more charge. Ask learners who miss an item to add it before the end of the lesson.
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D. TEACHER REFLECTION

After the lesson, reflect: Did most groups obtain readings close to 0, 30 and 50 degrees, and if not, was dampness or too few strokes the cause? Did learners keep to electrons as the only movers in the Explain phase? Which learners confused 'attraction' with 'detection' of charge and thought the leaf was pulled to the strip? Did the sign test make sense to learners, especially why a rise is a surer test than attraction? What did Amina's Mombasa question show about the class's understanding of humidity? Plan to use the evidence from this lesson, especially the sign test, earthing and the relationship between strokes and angle, as the starting point for Lesson 9's applications and final explanation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Record what you observed with the gold-leaf electroscope: the leaf at 0 degrees when neutral; about 30 degrees with a negative polythene strip (20 strokes); the leaf staying up after contact; about 50 degrees with 40 strokes; the angle going above 30 degrees with a negative object and below 30 degrees with a positive object; and 0 degrees after earthing with a finger.
What did I learn?	Write what you learned: a gold-leaf electroscope detects charge because the leaf and stem carry the same sign and repel; a bigger leaf divergence shows more charge; an increase on a charged electroscope shows the same sign and a decrease shows the opposite sign; earthing lets electrons flow to or from the Earth and discharges the electroscope.
How does this explain the phenomenon?	Explain how this helps answer the Driving Question: the rubbed balloon is shown to be charged because it makes the leaf diverge, its negative sign can be tested, and the amount of charge relates to the number of strokes because more electrons are transferred, not created.

LESSON 9 (80 minutes (2 x 40-minute periods)): Applications and Consolidation: From the Mysterious Balloon to Lightning Conductors, and the Final Explanation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the ideas from Lessons 1 to 8 to explain how electrostatics is used and managed in technology, and then write a complete final explanation of the anchoring phenomenon (the rubbed balloon, hair, wall and paper) that answers the main driving question.
Knowledge	By the end of the lesson learners should be able to: (1) explain how a lightning conductor protects a building in Kericho or Kisumu using point discharge and earthing; (2) explain why a fuel tanker at a Nairobi petrol depot trails an earthing chain or bonding wire; (3) describe how a photocopier and an electrostatic precipitator (e.g. at Athi River cement factory) use attraction between unlike charges; (4) write a final explanation of the balloon phenomenon covering friction, transfer of electrons, signs of charge, hair repulsion, induction on the wall and paper, humidity, the 1 nC charge value and conservation of charge.
Skills	Developing and using models (revising the particle-level model of balloon, hair, wall and paper); constructing explanations; analysing and interpreting data (humidity data, 1 nC balloon charge, electroscope and distribution readings); obtaining, evaluating and communicating information; transferring ideas to new situations.
Attitudes	Curiosity about how the invisible forces around us are used and controlled; respect for safety in storms and near fuel; appreciation of the value of evidence; collaboration and open-mindedness when revising ideas; pride in Kenyan engineers and technology.
Key Inquiry Question	Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? (Also: How is electrostatics used in everyday life?)
Purpose in Storyline	This is the final lesson of the storyline. Lessons 1 to 8 gave learners the pieces: charge signs and the rule for attraction and repulsion (L2), friction and electron transfer (L3), conductors and insulators (L4), contact charging (L5), induction and earthing (L6), charge on outer surfaces and point discharge (L7), and detecting charge with the electroscope (L8). Here learners use L7 point discharge and L6 earthing to understand technology, then bring everything together, complete the DQB, compare their Lesson 1 model with their final model, and write the final explanation of the anchoring phenomenon.
Safety Notes	Mr Mwangi uses only dry, low-energy demonstrations with the balloon, polythene strip and electroscope. No high voltage equipment is used; applications are studied through diagrams, short video or images and role-play. Remind learners never to shelter under a lone tree or stand in open ground during a thunderstorm, and never to use phones near fuel or take part in fuel handling. Keep paper pieces off the floor to avoid slips.

B. LESSON OVERVIEW

Mr Mwangi opens by returning to the balloon: Otieno's balloon sticking to the wall, Wanjiru's rising hair, Amina's coastal home where it never works in the rainy season, and Kipchoge's question about whether the shock is like lightning. Learners first predict how engineers could use or control the same ideas in Kericho, Kisumu, Nairobi depots and Athi River. They then observe four application cases (lightning conductor, fuel tanker earthing, photocopier, electrostatic precipitator) using diagrams and a quick balloon-and-paper demonstration, and explain each with ideas from earlier lessons. In the second period learners complete the Driving Question Board, compare their Lesson 1 model with the final model, and individually write the Final Explanation answering: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? The lesson closes with a summary table of what was observed, learned and explained.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 07 Lightning Arrestors. Lightning Conductor Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: electrostatics lightning conductor point	Learners sit in groups of four and look at four images on the board: a lightning conductor on a building in Kericho tea country, a fuel tanker with a chain trailing at a Nairobi petrol depot, a school-office photocopier, and a tall chimney at a cement factory in Athi River. Mr Mwangi also holds up the rubbed balloon sticking to the wall from Lesson 1. Individually, learners write a prediction in their notebooks: (a) What do you think each device does with electric charges? (b) Which idea from our lessons might it use? Then they share with their group and agree one prediction per device, marking each as 'attract', 'earthing', 'point discharge' or 'not sure'. Kipchoge's question is posted: 'Is the shock from a jersey like lightning?' Learners predict yes or no and give a reason.	Say: 'Last week we learned how to detect and measure charge. Today we ask: where do these ideas matter in Kenya? Look at the four pictures. Write what you think is happening to the charges.' Give silent individual writing time, then WAIT TIME of 10-15 seconds after asking: 'Kipchoge asked if the shock from a jersey is like lightning. What do you predict?' Cold-call 4 learners, one per device, to give a group prediction, and 2 more for Kipchoge's question. Say: 'I am not looking for the right answer yet. I want your reasoning.' Record predictions on the board under the heading 'Predictions Before Applications'. Do not confirm or correct. Ask: 'Which of Lessons 1 to 8 will help us decide?' and list learners' answers.	Predict-Observe-Explain (prediction stage) with a Think-Pair-Share routine: committing to a prediction exposes the ideas learners already hold from Lessons 1 to 8 and creates a reason to examine the application evidence. Linking the images to the balloon shows that the same ideas are being transferred to technology.	Look for: learners using the terms 'electrons', 'earthing', 'attract' or 'sharp point' in predictions; learners who say charge is 'created' or that 'positive charge moves' (note for later correction through evidence); learners who link the jersey shock to lightning. Tally on a class sheet how many groups link the tanker to earthing and the lightning conductor to a sharp point. Aim: at least 6 of 10 groups give a reason, not only a guess.
Observe Phase  VIDEO: 07 Lightning Arrestors. Lightning Conductor Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: electrostatics	Station-style evidence gathering in groups (about 10 minutes per station, 4 stations, rotating, with data cards). Station 1 Lightning conductor (Kericho and Kisumu): a diagram of a pointed metal rod on a building joined by thick copper strip to a buried earth plate. Learners recall the Lesson 7 pointed-conductor readings (pointed end about 45 degrees, flat end about 20 degrees) and record what this means for where charge crowds. Station 2 Fuel tanker at a Nairobi petrol depot: a diagram of a tanker with an earthing chain or bonding wire. Learners note that fuel flowing in the pipe rubs on the pipe wall and charge can build up	Circulate and ask at each station: 'What is the charge doing here? Where is it going?' At Station 1 ask: 'Looking at the pointed end reading from Lesson 7, why is a pointed rod used?' At Station 2 ask: 'What does the chain give the electrons a path to do?' WAIT TIME 10-15 seconds before accepting any answer. Prompt: 'Use the word electrons, not charge moving, and not static electricity as an explanation.' Cold-call 2 learners at each station for a one-sentence observation before rotating. At Station 3 say: 'Keep the strip dry. What do the paper pieces do and why do some fall off again?' Prompt with Lesson 5	Gallery-walk station analysis with data cards (Obtaining, evaluating and communicating information): moving through four contexts lets learners see the same small set of ideas (unlike charges attract, electrons transfer, point discharge, earthing) recur in different technologies, which builds transfer and consolidates the model rather than adding new facts.	Check data cards for: Station 1 - charge crowds at the point, strong field, charge leaks away through the air, electrons flow through the copper strip to earth; Station 2 - chain lets electrons flow to ground, so no charge builds up and no spark; Station 3 - toner and drum have opposite signs so attract; Station 4 - dust given a charge, attracted to plates of opposite sign. Observable indicator: learners correctly use 'electrons' in at least 3 of 4 cards. Note groups who say 'static electricity' as the explanation and prompt them to name the transfer.

lightning conductor point	on the tank; they list what an uncontrolled spark could do near fuel vapour. Station 3 Photocopier at a school office and Station 4 Electrostatic precipitator at Athi River: flow-chart cards show a charged drum attracting oppositely charged powder (toner), and smoke dust particles being given negative charge and attracted to positively charged plates. At Station 3 learners also do a quick dry demonstration: pass a rubbed polythene strip over small paper pieces (about 1 cm x 1 cm) on a desk and observe them jump and stick, like toner on the drum. They record what they see, then sketch the charge signs on a data card. They also revisit the balloon humidity data: about 3 minutes on a wall at 35% relative humidity in Nairobi; under 30 seconds or none at about 80% in Mombasa.	contact idea if needed. At Station 4 ask: 'Which charges attract: like or unlike? How do you know from Lesson 2?' Watch time and keep groups on schedule. Make sure all data cards are completed before the Explain Phase.		
Explain Phase  VIDEO: 07 Lightning Arrestors. Lightning Conductor Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: electrostatics lightning conductor point	Learners return to their groups and write a short claim-evidence-reasoning (CER) explanation for each application, then compare with their Predict Phase table. Lightning conductor: claim - it protects a building; evidence - pointed end gave a larger leaf reading (about 45 degrees) than the flat end (about 20 degrees); reasoning - charge crowds at points, the strong field ionises the air, charge leaks away gradually, and the thick copper strip and earth plate let electrons flow safely to or from the ground, which is why lightning conductors are used on buildings in Kericho and Kisumu. Fuel tanker: claim - the chain prevents sparks; reasoning - earthing lets electrons flow to the ground so charge from friction cannot build up (Lesson 6	Say: 'Choose one application. Make a claim, give evidence from our lessons, and give reasoning using electrons.' Give a sentence frame on the board: 'I think ____ because the data show ____ which means electrons ____.' WAIT TIME 10-15 seconds after each question. Cold-call 5 learners to share, then ask 3 others: 'Do you agree or disagree with Wanjiru's group? What evidence supports that?' For Kipchoge's question say: 'Both are discharge of built-up charge. What is very different about the amount of charge?' Guide: the jersey shock involves a tiny charge (like the balloon's about 1 nC) while lightning involves vastly more. Correct misconceptions: 'Charges are not created, they are transferred.' and 'Positive charge does not move in	Claim-Evidence-Reasoning (CER) with a sequencing chain: CER forces learners to link evidence from earlier lessons to a mechanism, while the cold-call story chain builds the complete causal sequence of the balloon phenomenon collaboratively. Comparing the jersey shock and lightning lets learners use the same mechanism at different scales, deepening understanding.	Observable indicators: a learner states that the lightning conductor works by point discharge and earthing, not by 'attracting lightning'; a learner states that the earthing chain lets electrons flow to ground; a learner uses 'transfer of electrons' and 'conserved' in the story chain; a learner explains the wall and paper attraction as induction (near side opposite charge, attraction stronger than repulsion). Count: aim for at least 80% of learners to contribute a correct step in the chain. Note common errors (for example, saying the balloon 'makes' charge or that neutral objects are repelled) and record them for the final check.

	earthing). Photocopier and precipitator: unlike charges attract. Then each group answers Kipchoge's question: how is the shock from a woollen jersey in Nairobi like and unlike lightning? Learners then do a whole-class share-out. Finally they begin the Final Explanation skeleton in pairs: they list the steps of the balloon story in order (friction, transfer of electrons, balloon negative and hair positive, hair repels itself and is attracted to the balloon, induction in wall and paper, humidity and slow leakage, 1 nC about 6×10^9 electrons out of about 10^{23} atoms, conservation of charge).	a solid, electrons do.' Then say: 'Now, using the balloon, tell me the story from first rubbing to the wall.' Use a cold-call chain of 6 learners, each adding one step, while you write the steps on the board. Challenge: 'Amina, why did the balloon fail at your coastal home?' WAIT TIME 10-15 seconds, then connect to humidity: at about 80% relative humidity the balloon sticks under 30 seconds or not at all because water vapour lets the charge escape.		
Driving Question Board (DQB) Creation  VIDEO: 07 Lightning Arrestors, Lightning Conductor Source: Seavuria  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: electrostatics lightning conductor point	Learners gather at the Driving Question Board that has been built since Lesson 1. In groups they take the sticky notes of questions and sort them into 'answered with evidence', 'partly answered' and 'still unanswered'. For each answered question they write a short answer card with the lesson number and evidence (for example: 'Where do charges come from? Electrons transfer from hair to balloon in friction, L3, balloon gains about 1 nC, about 6×10^9 electrons'). They match the seven KICD inquiry questions to lessons: charges (L2), conductors and insulators (L4), charging conductors (L5 and L6), where charge goes (L7), detecting and measuring (L8), uses (L9). Each learner adds one new question of their own that goes beyond the unit (for example, how is the charge in a thundercloud separated?) onto a 'Next questions' area. The class reads the main driving question aloud	Say: 'Our DQB has grown since Lesson 1. Which questions can we now answer with evidence?' Hand each group a bundle of notes and WAIT TIME of 10-15 seconds before they start sorting. Ask: 'What evidence shows this is answered, not just guessed?' Cold-call 5 groups to read one answer card, each from a different lesson. Say: 'Which ones are only partly answered? What would we need to find out?' Accept honest unknowns, for example, why exactly the electrons move from hair to latex (latex holds them more strongly). Ask: 'What new question do you have now?' and collect at least 8 on the board. Praise the quality of questions: 'That is a scientist's question.' Photograph or note the final DQB for the class record.	Driving Question Board (DQB) review and sort: revisiting student-generated questions shows learners their own growth in understanding and makes the link between each lesson and the anchoring phenomenon visible. Sorting by level of evidence builds the habit of grounding claims in data, and the 'next questions' area models how science generates new inquiry.	Observable indicators: groups correctly place at least 5 of the 7 KICD questions with a lesson and a piece of evidence; the answer cards use the conventions (electrons, transfer, earthing, discharge); at least 8 new, relevant questions are produced; learners can state which questions remain open and why. Check for a learner who still files a question under 'answered' without evidence and ask a probing question.

	and decides together that it has been answered.			
Model Building Phase  VIDEO: 07 Lightning Arrestors, Lightning Conductor Source: Seavuria  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: electrostatics lightning conductor point	Learners take out the model they drew in Lesson 1 (their initial ideas of the balloon, hair, wall and paper) and the revised versions from earlier lessons. On a fresh page they draw the final labelled particle-level model showing four parts: (1) balloon and hair before rubbing: neutral, with equal positive and negative charges; (2) after rubbing: electrons transferred from hair to latex, balloon negative (excess electrons, about 6×10^9 extra, around 1 nC), hair positive (deficit of electrons); (3) hair strands both positive repel each other and rise, and are attracted to the negative balloon; (4) wall and paper neutral, electrons shift slightly away from the near side, leaving the near surface positive, so attraction is stronger than repulsion; plus a note on humidity: at about 35% relative humidity charge leaks away slowly, at about 80% water vapour lets it escape. They add a small inset for one application of their choice (lightning conductor with point discharge and earthing, or tanker earthing). Then each learner writes the Final Explanation individually (8 to 10 sentences) answering: 'Where do electric charges come from? Why does a rubbed balloon attract or repel other objects?', using at least six key terms: friction, electrons, transfer, negative, positive, induction, repel, attract, earthing, humidity, conserved. Finally learners place their Lesson 1 model and the final model side by side and write three changes in a 'What changed in my thinking' box.	Say: 'Place your Lesson 1 model next to a blank page. Now draw what is really happening, at the level of electrons. Only electrons move in solids.' Provide a checklist on the board: labels for balloon, hair, wall, paper; signs on each; arrows showing electron transfer and electron shift; note on humidity; number about 6×10^9 electrons of about 10^{23} atoms. WAIT TIME 10-15 seconds before circulating. Ask: 'Why is the hair positive if it lost electrons? Why does the hair rise?' 'Why is the neutral paper attracted but not repelled?' Cold-call 3 learners to explain a part of their model to the class, and ask 2 others to critique: 'What would you add or change?' Remind them of the conventions: 'Charge is not created, it is transferred; total charge is conserved.' Walk around and check 8 to 10 learners' written explanations; give feedback with one star (strength) and one step (improvement). Close by asking 4 learners to read a key sentence of their Final Explanation, and ask the class to listen for the electron story. Invite Wanjiru, Otieno, Amina and Kipchoge's questions to be answered by name: 'Does your explanation answer Amina's coastal question? Kipchoge's lightning question?'	Cumulative model revision and model comparison (Developing and using models): placing the Lesson 1 model beside the final particle-level model makes conceptual change visible and gives learners ownership of their progress. Writing a Final Explanation with required key terms integrates all eight lessons into one causal account, consolidating the idea that the balloon phenomenon is explained by electron transfer, induction and leakage.	Use a four-point checklist for the Final Explanation and model: (1) friction transfers electrons from hair to latex (balloon negative, hair positive); (2) hair repulsion and attraction to balloon explained by like and unlike charges; (3) neutral wall and paper attracted by induction (near side opposite, attraction stronger than repulsion); (4) humidity and conservation of charge (including 1 nC about 6×10^9 electrons, tiny fraction of about 10^{23} atoms). Success: at least 3 of 4 points for most learners. Observable error indicators: 'charge was created', 'positive charge moved', 'static electricity' as the explanation, or 'neutral paper is repelled'. Record scores and plan follow-up for those with fewer than 3 points. The three-change reflection should identify at least one specific shift (for example, from 'magic' or 'magnetism' to electron transfer).

D. TEACHER REFLECTION

After the lesson, reflect: (1) Could most learners explain the lightning conductor using point discharge and earthing, and the tanker chain using earthing, rather than saying it 'attracts' lightning or 'removes static'? (2) In the Final Explanations, how many learners used electrons correctly and avoided saying charge is created or that positive charge moves in solids? (3) Did the model comparison show a clear shift from the Lesson 1 ideas, and which learners still hold the magnetism misconception? (4) Did Amina and Kipchoge's questions get answered with evidence (humidity data and the jersey shock versus lightning)? (5) Was WAIT TIME of 10-15 seconds used consistently, and did cold-calling spread across the class? (6) What follow-up is needed for learners who scored below 3 of 4 points: for example, a short re-teach of induction on the wall and paper, or of earthing? Record next steps and consider a short revision quiz in the following lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? (Examples: the rubbed balloon raises hair and sticks to the wall for about 3 minutes at about 35% humidity but only under 30 seconds at about 80%; paper pieces jump to a rubbed polythene strip; the lightning conductor has a pointed rod and a thick copper strip to an earth plate; a fuel tanker trails an earthing chain; a photocopier and a precipitator use attraction between unlike charges.)
What did I learn?	What did we learn? (Examples: charges are not created, electrons are transferred by friction and total charge is conserved; the balloon gains about 1 nC, about 6×10^9 electrons, out of about 10^{23} atoms; unlike charges attract and like charges repel; neutral objects are attracted by induction; charge crowds at points (point discharge) and earthing lets electrons flow to or from the ground; humidity lets charge leak away.)
How does this explain the phenomenon?	How can we explain the phenomenon? (Write your Final Explanation: when the balloon is rubbed on dry hair, electrons transfer from hair to latex because latex holds electrons more strongly, so the balloon is negative and the hair positive; the positive hair strands repel each other and rise and are attracted to the balloon; the negative balloon induces opposite charge on the near side of the neutral wall and paper so they attract; on dry days charge leaks slowly but on humid days water vapour lets it escape; technology such as lightning conductors, tanker earthing, photocopiers and precipitators uses the same ideas.)

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.